# Bias Reduction in TSTSLS Applications: A Quantile Adjustment Method

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**Abstract**

Many research questions require income and a second variable measured for the same people, but the two are often recorded in separate samples. Two-sample two-stage (TSTS) imputation bridges the samples by predicting income from characteristics both record, however, the predictions are less dispersed than actual income, and they miss any link between income and the second variable that the shared characteristics do not capture. Existing corrections fix these problems one at a time, and fixing only the first can make an estimate worse. I propose a quantile adjustment that replaces each prediction with the observed income at the same rank. It restores the income distribution and, in the same step, shifts the covariance with the second variable. I characterise when this shift helps and provide two diagnostics. In applications to Mexico, Ecuador and Chile, the adjustment comes closest to the full-sample benchmark on 14 of 19 estimands and within 0.6 percentage points of the best method on the other 5.

**Keywords:** Two-sample two-stage imputation, Income distributions, Quantile adjustment, Inequality of opportunity, Intergenerational mobility.

**JEL Classification:** C81, C83, D31

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# 1 Introduction

A number of research questions require income measured alongside a second variable. Welfare analysis needs income next to consumption; mobility research needs a parent's earnings next to their child's; inequality of opportunity needs income next to the circumstances a person was born into. In many cases, no single source observes both, so these pairs are rarely recorded together and often cannot be. Since the second variable lies outside the source where income is observed, I call it the *external variable* throughout.

The standard way to combine two such datasets is survey-to-survey imputation through the two-sample two-stage (TSTS) procedure (Klevmarken 1982; Inoue and Solon 2010), which uses the characteristics both record, the *common predictors*, to build a statistical bridge between them. This bridge distorts income measures in two ways. Predicted income is less dispersed than observed income, and it misses any link between income and the external variable that the common predictors do not capture. Existing corrections are tailored to a particular estimand and address the two distortions one at a time. Fixing only one can even increase the bias of the final estimate.[1]

In this paper I show that researchers do not have to choose which distortion to correct. I propose a quantile adjustment that replaces each prediction with the observed income at the same rank. It restores the full income distribution and, in the same step, shifts the covariance with the external variable toward its true value when the common predictors and the unexplained part of income relate to that variable in the same direction. I characterise when it helps, provide two diagnostics, and show in three applications that it comes closest to the true value on 14 of the 19 estimands I report.

The TSTS procedure works in two stages. In the first, the researcher fits a model of income on the common predictors in a *donor sample*, where income is observed, and uses it to predict

[1] Methods that borrow values from other respondents in the same sample, such as predictive mean matching (Little 1988), hot-deck imputation (Rubin 1986) and multiple imputation (Rubin 1987), need some respondents to report income. Income is missing from every row of the target sample, so none of them applies here.

income in a *target sample*, where it is missing. In the second, she estimates the relationship of interest with predicted income in place of income. The first stage is usually a log-linear regression, although machine learning methods have also been applied (Bloise et al. 2021; Viljanen et al. 2022). I call the use of these predictions without any correction *baseline imputation*.

Replacing income with its prediction introduces two distortions that the literature has documented. The first is *variance bias*. A predicted value is an average over people who share the same characteristics, so it smooths away everything the first stage cannot explain. Predicted incomes are therefore less spread out than observed incomes (see e.g. Rubin 1976; McKenzie 2005; Ferreira, Gignoux, and Aran 2011). In the applications that follow, for example, baseline imputation recovers between 48 and 65 per cent of the variance of log income.

The second is *covariance bias*, which Cortes Orihuela et al. (2024) call projection bias. Predicted income keeps only the part of its relationship with the external variable that runs through the common predictors. Consider two parents from different families who have the same education and occupation but earn different amounts. If the higher earner also tends to have higher-earning children, that link disappears once parental income is replaced by its prediction. In the same applications, baseline imputation recovers between 58 and 78 per cent of the covariance between income and the external variable.

Which bias matters most depends on the estimand, and in Table 1 I sort common applications accordingly. Statistics of income alone, such as poverty headcounts (Elbers et al. 2003; Sinha Roy and Weide 2022) and inequality indices (Ferreira, Gignoux, and Aran 2011; Betti et al. 2024), suffer only from variance bias. Correlations and elasticities (Björklund and Jäntti 1997; Mazumder 2005; Donatiello et al. 2025) suffer from both, with covariance bias in the numerator and variance bias in the denominator. Regression coefficients with income as the dependent variable, as in inequality of opportunity decompositions (Ferreira, Gignoux, and Aran 2011; Torres et al. 2025), suffer only from covariance bias, because the spread of income does not enter them.

Table 1: Which Bias Reaches Which Estimand

| Group | Estimand | Variance bias | Covariance bias |
|---|---|---|---|
| Marginal distribution | Poverty rates and headcounts | Present | Absent |
| | Inequality measures (Gini, MLD) | Present | Absent |
| | Synthetic panel statistics | Present | Absent |
| Joint distribution (A) | Intergenerational elasticity | Present | Present |
| | Income–consumption correlation | Present | Present |
| | Treatment effect estimates | Present | Present |
| | Productivity dispersion | Present | Present |
| Joint distribution (B) | IOp circumstance coefficients | Absent | Present |
| | House value–consumption regression | Absent | Present |

*Notes:* Present means that the bias reaches the estimand under baseline imputation. Variance bias reaches estimands that depend on the spread of income, and covariance bias reaches estimands that depend on its covariance with the external variable. Joint distribution (A) collects estimands in which income enters symmetrically with the external variable or as a regressor, and Joint distribution (B) those in which income is the dependent variable.

Existing corrections treat the two biases as separate problems. For variance bias, the usual fix is stochastic augmentation, which adds to each prediction a random draw from the donor residuals (Rubin 1976; McKenzie 2005; Ferreira, Gignoux, and Aran 2011; Dang 2021). This restores the variance but not necessarily the tails, to which many inequality indices are sensitive (Atkinson 1970; Cowell 2011; Betti et al. 2024), so poverty headcounts tend to come back more accurately than inequality measures (Demombynes et al. 2007). And because the random draws have nothing to do with the external variable, the covariance stays where baseline imputation left it.

Corrections for covariance bias act on the second-stage estimate rather than on the imputed values. Crossley et al. (2022) show that regression-based and matching-based imputation both give inconsistent second-stage estimates and propose rescaled regression prediction. Cortes Orihuela et al. (2024) combine the two corrections for the intergenerational elasticity, first rescaling the variance of imputed income and then applying the rescaled regression prediction of Crossley et al. (2022). Because these corrections target the estimand, each has to be worked out again for every estimand a researcher might care about.

Correcting one bias at a time is most risky for ratio estimands such as correlations and elasticities. There the two biases partly cancel, so removing only one leaves the other fully exposed.

The quantile adjustment I propose corrects the imputed values themselves rather than the estimate built from them. It keeps each observation's rank in the predicted income distribution but gives it the income found at that rank in the observed distribution. An observation at the 90th percentile of predicted income, for instance, receives the income at the 90th percentile of observed income in the donor sample.

The first consequence is that the adjusted values reproduce the distribution of observed income. Each adjusted value is an observed income quantile, so the spread, the tails and any inequality index computed from them all come back. Because the correction changes the imputed values rather than the estimand, the same adjusted incomes can feed any second-stage analysis, from a correlation to a regression with income on either side.

The second consequence is that the covariance with the external variable moves as well. On average, the adjustment raises incomes at the top of the predicted distribution and lowers them at the bottom. In the mobility example, the parents at the top are those whose education and occupation predict high earnings, and those same characteristics also predict high-earning children. The adjustment therefore raises parental incomes most where offspring incomes are high, which restores part of the link that baseline imputation lost.

How much of the link comes back depends on two relationships. I measure it by the recovery fraction, the share of the missing covariance that the adjustment puts back. The adjustment depends only on the common predictors, so it can supply only covariance that runs through them, and the missing covariance is the part that does not. The result therefore depends on how the common predictors relate to the external variable and on how the part of income they cannot explain relates to it.

When the two relationships point the same way, the adjustment moves the estimate toward its true value. In the mobility example, this is the case when parental background and

the links between parents and children that the survey does not record both raise offspring income (Björklund and Jäntti 1997). The adjustment then usually closes part of the gap. It can also overshoot and put back more covariance than was missing, but it still ends closer to the true value than baseline imputation as long as it puts back less than twice the missing amount.

When the two relationships point in opposite directions, the adjustment moves the estimate the wrong way. Suppose a researcher imputes husbands' earnings to study how they relate to their wives' earnings. Men whose characteristics predict high earnings tend to marry women who earn more (Eika et al. 2019), but within a group of similar men, wives tend to work less when their husbands earn more than predicted (Devereux 2004; Blau and Kahn 2007). These results rest mainly on the first stage being right on average for every combination of the common predictors.

The part of income the predictors miss is never observed, so I develop two diagnostics that a researcher can compute before imputing. The first is a screen that flags predictors which explain the external variable much better than they explain income. Such predictors push the adjustment toward overshooting, because they absorb covariance the first stage would otherwise have left for the adjustment to recover.

The second diagnostic is a threshold. It tells the researcher how strongly the unexplained part of income would have to be related to the external variable for the adjustment to recover the missing covariance exactly, which she can weigh against theory and prior evidence.

I evaluate the method in three applications that place predicted income in each position it can take in a second stage, using an income–consumption correlation in Mexico, an intergenerational elasticity in Ecuador and an inequality of opportunity decomposition in Chile. In each, income and the external variable are observed together, so splitting the data into a donor and a target sample recreates the two-sample problem, and the full-sample value serves as a benchmark. Repeating the split 1,000 times shows how much a single application can vary.

Across the applications, the quantile adjustment comes closest to the benchmark on 14 of the 19 estimands and stays within 0.6 percentage points of the best method on the other 5. Its gains are largest where a covariance or the tails of the distribution matter. It cuts the largest error in the Chilean circumstance coefficients from 44 to 24 per cent, brings the Ecuadorian elasticity within 3 per cent of the benchmark, against 21 per cent for baseline imputation, and brings total inequality and the inequality attributable to circumstances within about 4 per cent, where baseline imputation misses both by more than a third. Stochastic augmentation does best on the three income variances but ends furthest from the benchmark on all four ratio estimands.

In this paper I contribute to the literature on TSTS estimation (Klevmarken 1982; Inoue and Solon 2010; Crossley et al. 2022; Cortes Orihuela et al. 2024). Existing corrections act on a particular estimand and treat the two biases separately. The quantile adjustment acts on the imputed values, handles both biases in one step, and needs no derivation specific to each estimand. The analysis of the recovery fraction also shows when a correction of this kind helps and when it hurts.

I also contribute to the use of survey-to-survey imputation for distributional analysis (Elbers et al. 2003; Demombynes et al. 2007; Dang 2021; Sinha Roy and Weide 2022). Because the adjusted values reproduce the whole income distribution, inequality measures that depend on the tails come back without a separate correction. So that practitioners can apply the method directly, I developed the R package `tstsqa`, available at github.com/PedroToL/tstsqa, which implements the adjustment together with both diagnostics.

The rest of the paper is structured as follows. In Section 2 I set up the problem and define the two biases, and in Section 3 I develop the quantile adjustment and work out when it helps. In Section 4 I cover implementation and the diagnostics, in Section 5 I present the applications, and in Section 6 I discuss what they show. I conclude in Section 7.

# 2 Setup and Notation

Consider the problem of estimating the relationship between income $y$ and a variable of interest $z$ — such as consumption, wealth, or parental income — when the researcher has to rely on separate samples that observe these variables independently. Both samples contain a common set of predictors $\mathbf{X}$.

The *donor sample* consists of $n_d$ observations from the joint distribution of $(y, \mathbf{X})$, where $y$ denotes a measure of income and $\mathbf{X}$ is a vector of covariates, and $z$ is unobserved. The *target sample* consists of $n_t$ observations from the joint distribution of $(z, \mathbf{X})$, which records $z$ but not $y$. Both samples are assumed to be random draws from the same population distribution, and that $\mathbf{X}$ contains meaningful predictors of $y$, so that the conditional expectation of $y$ given $\mathbf{X}$ varies with $\mathbf{X}$.

For example, in the intergenerational elasticity the target survey records offspring income $z$ alongside parental background $\mathbf{X}$, but not the parental income $y$ the elasticity needs. A second survey observes $y$ together with the same background variables and serves as the donor. The researcher estimates income on those common predictors in the donor survey, then applies the fitted model in the target survey to impute parental income for each offspring.

The goal is to recover properties of the joint distribution of $(y, z)$ by combining information from the two samples through a two-sample two-stage (TSTS) imputation procedure. In Sections 2 and 3 I characterise the biases at the population level, and I denote population quantities without subscripts.

## 2.1 First-Stage Imputation

At the population level, income decomposes as

$$y = f(\mathbf{X}) + \varepsilon, \tag{1}$$

where $f(\mathbf{X}) := \mathbb{E}[y \mid \mathbf{X}]$ and $\varepsilon := y - f(\mathbf{X})$, so that $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$ by construction.

The imputed outcome is the population prediction,

$$\hat{y} = f(\mathbf{X}). \tag{2}$$

In practice $f$ is unknown and the researcher assumes a functional form, most commonly log-linear, and estimates it on the donor sample, yielding $\hat{f}$ with $\hat{f} \xrightarrow{p} f$ under standard regularity conditions.

This imputation step induces a Berkson measurement error structure (Berkson 1950; Bound et al. 2001; Crossley et al. 2022). Under classical measurement error the error is independent of the true variable. Under Berkson error the error is instead mean-independent of the measured variable. Here $\hat{y}$ is the measured variable and $\varepsilon$ the prediction error, so that

$$y = \hat{y} + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid \hat{y}] = 0, \tag{3}$$

where the second condition follows from $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$ and the fact that $\hat{y}$ is a deterministic function of $\mathbf{X}$.

The prediction error satisfies $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$ by construction, but it need not satisfy $\mathbb{E}[\varepsilon \mid z] = 0$. Some of the variation in $y$ that $\mathbf{X}$ leaves unexplained can be correlated with $z$, and the imputation model then leaves out information about how $y$ and $z$ move together.

The choice of functional form for $\hat{f}(\cdot)$ adds a further complication. When the researcher estimates the first stage in log-linear form but the estimand is defined in levels, she has to retransform the predicted outcome (Manning 1998; Manning and Mullahy 2001). A well-known concern is that $\mathbb{E}[y \mid \mathbf{X}] \neq \exp(\mathbb{E}[\log y \mid \mathbf{X}])$, a consequence of Jensen's inequality. The smearing retransformation of Duan (1983) corrects for this by multiplying predictions by the average of the exponentiated donor residuals. That restores the mean in levels, and it restores the conditional mean as well when the log errors are homoskedastic. I discuss the heteroskedastic case in Appendix D.1.

## 2.2 Two Sources of Bias

The imputation procedure produces two sources of bias.

*Variance bias* arises mechanically from imperfect first-stage prediction. Because $\hat{y}$ is a conditional expectation, the law of total variance gives

$$\mathrm{Var}(y) = \mathrm{Var}(\hat{y}) + \mathrm{Var}(\varepsilon). \tag{4}$$

Write $R^2 := \mathrm{Var}(\hat{y})/\mathrm{Var}(y)$ for the share of the variance of $y$ that $\mathbf{X}$ explains, and assume $0 < R^2 < 1$ throughout. Then $\mathrm{Var}(\hat{y}) = R^2\,\mathrm{Var}(y) < \mathrm{Var}(y)$, so the imputed distribution is compressed relative to the distribution of $y$ by a factor that depends on the first-stage fit.

*Covariance bias*, which Cortes Orihuela et al. (2024) call projection bias, arises from the Berkson-style decomposition of $y$. The covariance between $y$ and $z$ decomposes as

$$\mathrm{Cov}(y, z) = \mathrm{Cov}(\hat{y}, z) + \mathrm{Cov}(\varepsilon, z), \tag{5}$$

where the first term captures the $y$–$z$ relationship that runs through $\mathbf{X}$ and the second the covariance the imputation step discards. TSTS procedures substitute $\hat{y}$ for $y$, so they recover only $\mathrm{Cov}(\hat{y}, z)$ and omit $\mathrm{Cov}(\varepsilon, z)$ entirely. When $\mathrm{Cov}(\varepsilon, z) > 0$, the imputation understates the true covariance between $y$ and $z$, and when $\mathrm{Cov}(\varepsilon, z) < 0$, it overstates it.

I call the unadjusted use of $\hat{y}$ in place of $y$ baseline imputation because both stochastic augmentation and the quantile adjustment build on it. Baseline imputation recovers $\mathrm{Var}(\hat{y}) = R^2 \cdot \mathrm{Var}(y)$ and $\mathrm{Cov}(\hat{y}, z) = \mathrm{Cov}(y, z) - \mathrm{Cov}(\varepsilon, z)$, distorting both the variance of income and its covariance with $z$.

Variance bias affects statistics of the marginal distribution of $y$ alone, such as its variance or inequality indices. Covariance bias affects statistics of the joint distribution of $y$ and $z$, such as their correlation or regression coefficients. When both biases are present, they can reinforce or offset one another. In Section 5 I trace how these distortions carry through to

specific estimands.

# 3 Addressing the Biases

Existing approaches address the two biases separately. To correct variance bias, researchers typically use stochastic augmentation, adding draws from the residual distribution to the prediction (Rubin 1976; Little and Rubin 2002).

To correct covariance bias, Crossley et al. (2022) show that regression-prediction and matching-based imputation both yield inconsistent second-stage estimates, and propose rescaled regression prediction, a two-step estimator that scales the second-stage coefficient to recover consistency. Cortes Orihuela et al. (2024) combine the two corrections for the intergenerational elasticity, first rescaling the variance of imputed income through stochastic augmentation and then applying the rescaled regression prediction.

Both corrections act on the estimand, so the researcher has to derive each for the estimand at hand. The quantile adjustment I develop below acts on the imputed values instead. It corrects both moments at once and leaves the second stage untouched.

## 3.1 The Stochastic Augmentation Method

Under stochastic augmentation, the researcher adds to the predicted value a random draw from the empirical distribution of first-stage residuals,

$$\tilde{y}_s = \hat{y} + \tilde{\varepsilon}, \tag{6}$$

where $\tilde{\varepsilon}$ is drawn from the distribution of $\varepsilon$ with $\tilde{\varepsilon} \perp\!\!\!\perp \mathbf{X}$. By construction, this restores the unconditional variance,

$$\mathrm{Var}(\tilde{y}_s) = \mathrm{Var}(\hat{y}) + \mathrm{Var}(\varepsilon) = \mathrm{Var}(y). \tag{7}$$

Because the draws are independent of $\hat{y}$ and $\mathbf{X}$, stochastic augmentation need not recover higher moments or the full shape of the outcome distribution.[2] Inequality indices such as the Gini coefficient and the mean log deviation are nonlinear functionals of the entire distribution and are often sensitive to tail behaviour (Atkinson 1970; Cowell 2011; Betti et al. 2024). Bias in these measures can therefore persist even after stochastic augmentation has matched the variance.

Drawing $\tilde{\varepsilon}$ from the distribution of $\varepsilon$ and exponentiating implicitly performs Duan's smearing retransformation. It therefore corrects the conditional mean in levels under the same condition as smearing, homoskedastic log errors, without further adjustment. The variance restoration of Equation (7) does not in general survive exponentiation, since exp is nonlinear.

For covariance bias, since $\tilde{\varepsilon}$ is drawn independently of $z$, $\mathrm{Cov}(\tilde{\varepsilon}, z) = 0$, and therefore

$$\mathrm{Cov}(\tilde{y}_s, z) = \mathrm{Cov}(\hat{y}, z). \tag{8}$$

Stochastic augmentation provides no improvement over baseline imputation for the covariance between imputed income and $z$.

## 3.2 The Quantile Adjustment Approach

The quantile adjustment applies a correction that varies with an observation's rank in the predicted distribution. One correction addresses both biases, restoring the marginal distribution of $y$ and moving its covariance with $z$.

The argument runs in four steps. In Section 3.2.1 I show that the adjustment restores the whole distribution of $y$. In Section 3.2.2 I show that it shifts the imputed covariance with $z$ by a single term, and that only the part of $z$ the predictors explain contributes to that

[2] Formally, if $\tilde{\varepsilon} \overset{d}{=} \varepsilon$ and $\tilde{\varepsilon} \perp\!\!\!\perp \mathbf{X}$, then $\mathrm{Var}(\tilde{y}_s) = \mathrm{Var}(y)$. The difference between their third central moments is $3\,\mathrm{Cov}(\hat{y}, \varepsilon^2)$, which can be nonzero when the conditional residual variance varies with $\hat{y}$, that is, under heteroskedasticity linked to the predicted level, and vanishes under homoskedasticity. If $\varepsilon$ itself is independent of $\mathbf{X}$, $\tilde{y}_s$ has exactly the distribution of $y$, so the shortfalls described in this section arise only when the spread or shape of $\varepsilon$ varies with $\mathbf{X}$.

shift. In Section 3.2.3 I compare the shift with the covariance baseline imputation omits, which gives the recovery fraction $\lambda$ and three regimes. In Section 3.2.4 I extend the results to several external variables.

Let $Q_{\hat{y}}(p)$ denote the quantile function of $\hat{y}$ for $p \in [0,1]$, and similarly $Q_y(p)$ for the observed outcome. Define the quantile-gap function

$$\eta(p) := Q_y(p) - Q_{\hat{y}}(p), \tag{9}$$

which measures the difference between the observed and predicted quantile functions at rank $p$. I adjust the imputed values as

$$\tilde{y}_q = \hat{y} + \tilde{\eta}, \tag{10}$$

where $\tilde{\eta} := \eta(p)$ and $p = F_{\hat{y}}(\hat{y})$ is the observation's own predicted rank. Each observation therefore receives the quantile gap evaluated at its own position in the predicted distribution.

In Figure 1 I show how to construct the quantile gap in an illustrative case. Because the predicted distribution is compressed relative to the observed one (Section 2.2), $Q_{\hat{y}}$ tends to lie above $Q_y$ at low $p$ and below it at high $p$, so $\eta(p)$ tends to be negative in the lower tail and positive in the upper.

### 3.2.1 Variance Bias Correction

With $p = F_{\hat{y}}(\hat{y})$ an observation's own predicted rank, $\hat{y} = Q_{\hat{y}}(p)$, so

$$\tilde{y}_q = Q_{\hat{y}}(p) + \eta(p) = Q_y(p). \tag{11}$$

The adjustment replaces each predicted value by the observed quantile at the same rank. When $\hat{y}$ is continuous, $p$ is uniform and $\tilde{y}_q$ has the distribution of $y$ exactly, including its higher moments and tail behaviour. Mass points in $\hat{y}$ break this step, and I discuss how to handle them in Section 4.3.

Exponentiation leaves ranks unchanged, and the quantile at a given rank of $\exp(y)$ is the

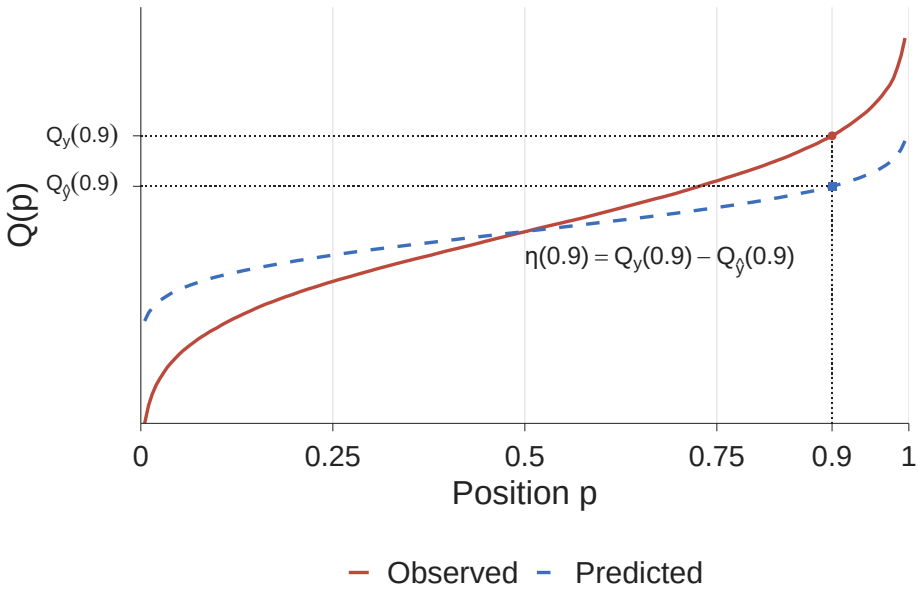



Figure 1: Construction of the Quantile Gap

*Notes:* Illustrative example using simulated data. The red solid line plots the observed quantile function $Q_y(p)$, and the blue dashed line plots the predicted quantile function $Q_{\hat{y}}(p)$. The vertical distance between the two lines at any rank is the quantile gap $\eta(p)$, and the marker at $p = 0.9$ shows one such gap, $\eta(0.9) = Q_y(0.9) - Q_{\hat{y}}(0.9)$.

exponential of the quantile of $y$. The adjusted values computed in logs and then exponentiated therefore coincide with those computed in levels, and no retransformation correction arises at any point.

### 3.2.2 Covariance Bias Correction

Since $\tilde{y}_q = \hat{y} + \tilde{\eta}$, linearity of covariance gives

$$\text{Cov}(\tilde{y}_q, z) = \text{Cov}(\hat{y}, z) + \text{Cov}(\tilde{\eta}, z), \tag{12}$$

so the adjustment shifts the imputed covariance by $\text{Cov}(\tilde{\eta}, z)$. That shift is not the prediction error put back. An observation's predicted rank $p$ determines its adjustment $\tilde{\eta}$, while the prediction error $\varepsilon = y - \hat{y}$ belongs to the individual. I show the difference in Figure 2. By Equation (3), $\varepsilon$ averages zero at every predicted rank, although its spread may vary with $p$.

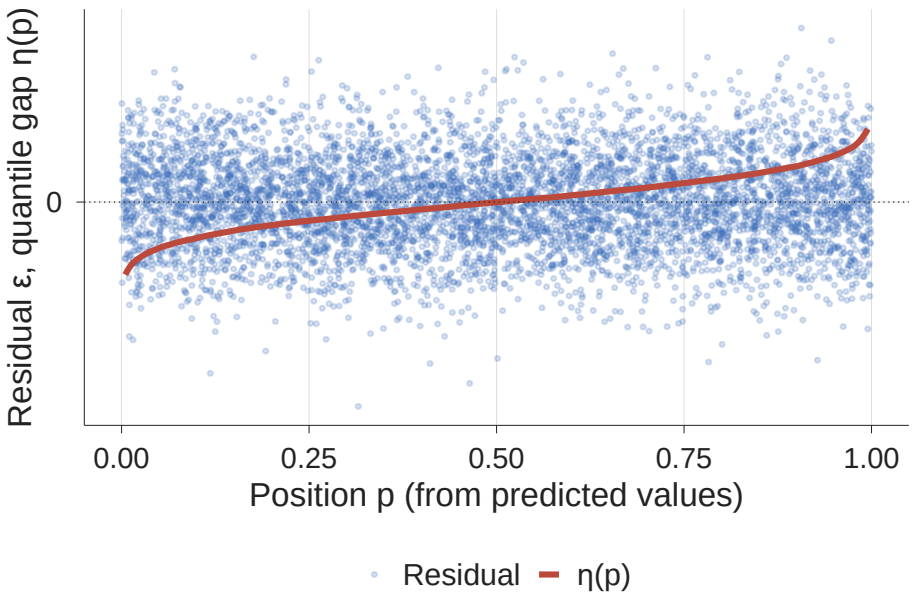



Figure 2: Adjustment and Prediction Error by Rank

*Notes:* Illustrative example using simulated data. Each point plots an individual first-stage residual $\varepsilon$ against its predicted rank $p = F_{\hat{y}}(\hat{y})$, and the solid line plots $\eta(p)$ on the same scale. The residuals average zero at each rank, whereas every observation at a given rank receives the single adjustment $\eta(p)$.

The poorest share of the population by predicted income is, on average, as well off as predicted, while the poorest share by true income can only be worse off (Appendix A). On balance the adjustment therefore lowers incomes at the bottom and raises them at the top, so

$$\mathrm{Cov}(\hat{y}, \tilde{\eta}) \geq 0. \tag{13}$$

The quantile gap need not rise at every rank for this to hold. In the Ecuadorian application of Section 5.2 it falls over part of the lower half of the distribution (Appendix Figure E.6), and the co-movement is still positive.

When $\mathrm{Cov}(\varepsilon, z) \neq 0$, define the recovery fraction

$$\lambda := \frac{\mathrm{Cov}(\tilde{\eta}, z)}{\mathrm{Cov}(\varepsilon, z)}, \tag{14}$$

so that

$$\mathrm{Cov}(\tilde{y}_q, z) = \mathrm{Cov}(\hat{y}, z) + \lambda \,\mathrm{Cov}(\varepsilon, z). \tag{15}$$

In words, $\lambda$ is the fraction of the omitted covariance $\mathrm{Cov}(\varepsilon, z)$ that the quantile adjustment puts back.

This fraction can take any sign and any magnitude. Whether the shift improves on baseline imputation therefore turns on how $z$ relates to $\tilde{\eta}$, so I split $z$ into the part the predictors explain and the part they do not. Define $g(\mathbf{X}) := \mathbb{E}[z \mid \mathbf{X}]$, so that

$$z = g(\mathbf{X}) + z_\perp, \qquad \mathbb{E}[z_\perp \mid \mathbf{X}] = 0. \tag{16}$$

Return to the intergenerational example. Here $g(\mathbf{X})$ is the offspring's expected income given parental background alone. It captures channels such as educational transmission, since parental education predicts offspring income partly through the offspring's own schooling.

Decomposing $\mathrm{Cov}(\tilde{\eta}, z)$ through Equation (16) identifies two channels,

$$\mathrm{Cov}(\tilde{\eta}, z) = \underbrace{\mathrm{Cov}(\tilde{\eta}, g(\mathbf{X}))}_{\text{predictor channel}} + \underbrace{\mathrm{Cov}(\tilde{\eta}, z_\perp)}_{\text{error channel}}. \tag{17}$$

Given that $\tilde{\eta}$ depends on $\mathbf{X}$ only through the rank of $\hat{y}$, and $z_\perp$ is uncorrelated with any function of $\mathbf{X}$ by construction, the error channel is exactly zero. The adjustment therefore cannot reach the component of $z$ that the predictors do not explain, which is the component baseline imputation discards. It follows that

$$\mathrm{Cov}(\tilde{\eta}, z) = \mathrm{Cov}(\tilde{\eta}, g(\mathbf{X})), \qquad \lambda = \frac{\mathrm{Cov}(\tilde{\eta}, g(\mathbf{X}))}{\mathrm{Cov}(\varepsilon, z)}. \tag{18}$$

Since $\tilde{\eta}$ depends on $\mathbf{X}$ through $\hat{y}$, only the variation in $g(\mathbf{X})$ that tracks predicted income contributes. Writing

$$\bar{g}(t) := \mathbb{E}[g(\mathbf{X}) \mid \hat{y} = t], \tag{19}$$

$\bar{g}(t)$ is the expected value of $z$ among observations sharing predicted income $t$. In the intergenerational example, $\bar{g}(t)$ traces how expected offspring income moves as predicted parental income $t$ rises. If parents with more education tend to have both higher income and higher-earning children, $\bar{g}$ is increasing in $t$. The predictor channel reduces to

$$\text{Cov}(\tilde{\eta}, g(\mathbf{X})) = \text{Cov}\left(\tilde{\eta}, \bar{g}(\hat{y})\right). \tag{20}$$

When $\bar{g}$ is nondecreasing, the adjustment and the expected external variable tend to rise together with predicted income, and the channel cannot be negative,

$$\bar{g} \text{ nondecreasing in } \hat{y} \implies \text{Cov}(\tilde{\eta}, g(\mathbf{X})) \geq 0. \tag{21}$$

The inequality is strict when $\bar{g}$ is strictly increasing, and the sign reverses when $\bar{g}$ is decreasing (Appendix A). I illustrate the increasing case in Figure 3, and in Appendix B I give conditions under which the sign can be read from regression coefficients.

### 3.2.3 The Sign and Magnitude of $\lambda$

Both covariances in Equation (14) simplify, and each keeps a different part of $z$. The numerator reduces to the predictor channel, by Equation (18). The denominator reduces to the error channel. Since $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$, the error is uncorrelated with $g(\mathbf{X})$, so

$$\text{Cov}(\varepsilon, z) = \text{Cov}(\varepsilon, z_{\perp}). \tag{22}$$

The adjustment supplies covariance from the part of $z$ the predictors explain, to replace covariance lost from the part they do not.

Writing each covariance as a correlation times two standard deviations,

$$\lambda = \frac{\kappa}{\rho} \cdot \frac{\sigma_{\tilde{\eta}}}{\sigma_{\varepsilon}} \cdot \frac{\sigma_{g(\mathbf{X})}}{\sigma_{z_{\perp}}}, \tag{23}$$

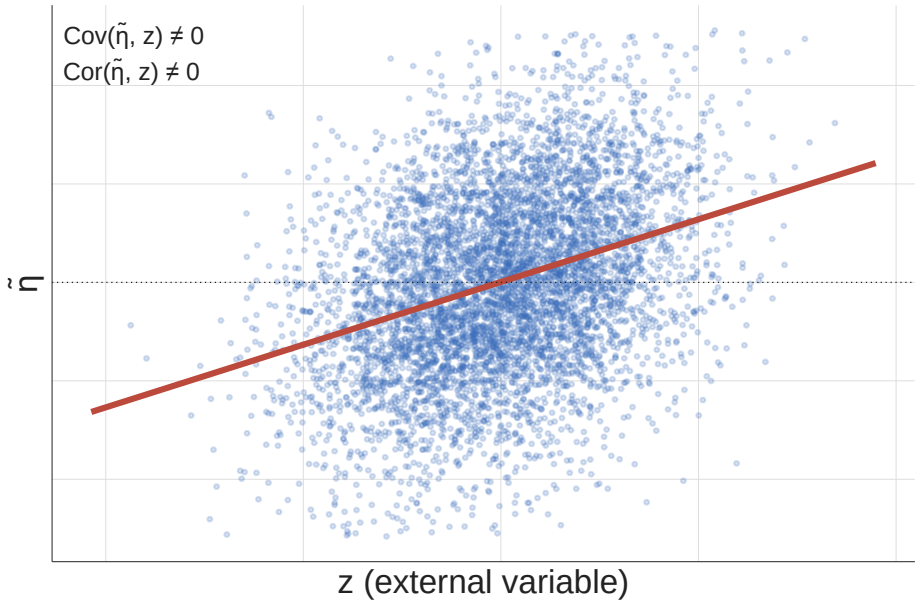



Figure 3: Adjustment and the External Variable

*Notes:* Illustrative example using simulated data. Each point plots an observation's realised quantile-gap adjustment $\tilde{\eta}$ against the external variable $z$, and the solid line is a linear fit. An upward-sloping line corresponds to a positive predictor channel.

where $\kappa := \text{Corr}(\tilde{\eta}, g(\mathbf{X}))$ and $\rho := \text{Corr}(\varepsilon, z_\perp)$. The first factor compares how well the adjustment tracks the predictable part of $z$ against how strongly income and $z$ move together outside the predictors. The second compares the size of the adjustment against the size of the error it stands in for. The third compares how much of $z$ the predictors explain against how much they leave behind.

Three regimes follow, together with the boundary case of exact recovery.

**Regime 1. Partial covariance adjustment** $(0 < \lambda < 1)$. The predictors and the channels they miss pull income and $z$ in the same direction, and the adjustment supplies less covariance than the first stage omitted.

In the intergenerational example, parental education predicts higher offspring income, and whatever else links parent to child does too. The adjustment moves the imputed covariance toward the truth without reaching it, and the residual bias keeps the sign of the baseline bias, attenuated by the factor $1 - \lambda$.

**Exact recovery** ($\lambda = 1$). The adjustment supplies exactly the covariance the first stage omitted, and the imputed covariance equals the truth.

**Regime 2. Over-recovery** ($\lambda > 1$). The two still point the same way, but the adjustment supplies more covariance than the first stage omitted. The imputed covariance then crosses the truth, and the residual bias takes the opposite sign to the baseline bias.

Predictors that are strongly related to $z$ but add little to the prediction of income produce that configuration. Adding such a predictor raises $\mathrm{Cov}(\widehat{y}, z)$ and shrinks the omitted covariance $\mathrm{Cov}(\varepsilon, z)$, while the adjustment inherits part of the added covariance through $\tilde{\eta}$.

**Regime 3. Mis-recovery** ($\lambda < 0$). The predictors point one way and the channels they miss point the other. The adjustment then supplies covariance of the wrong sign, moving the imputed covariance away from the truth and enlarging the baseline bias by the factor $1 + |\lambda|$.

In the intergenerational example this would require parental background to predict higher offspring income while the unmeasured links between parent and child predict lower. I give an example from spouses' earnings in Appendix B.

At $\lambda = 0$ the adjustment supplies no covariance at all and leaves the baseline value unchanged. This happens when $\bar{g}$ is flat, so that the expected external variable does not vary with predicted income, as when $z$ is independent of $\mathbf{X}$.

Subtracting $\mathrm{Cov}(y, z)$ from Equation (15) leaves a residual bias of $(\lambda - 1)\,\mathrm{Cov}(\varepsilon, z)$, against a baseline bias of $-\mathrm{Cov}(\varepsilon, z)$, so

$$\frac{\big|\,\mathrm{Cov}(\tilde{y}_q, z) - \mathrm{Cov}(y, z)\big|}{\big|\,\mathrm{Cov}(\widehat{y}, z) - \mathrm{Cov}(y, z)\big|} = |1 - \lambda|. \tag{24}$$

The adjustment reduces the magnitude of the bias when $|1 - \lambda| < 1$, that is, for $0 < \lambda < 2$. Over-recovery up to $\lambda = 2$ still leaves the estimate closer to the truth than baseline imputation. At $\lambda = 2$ the two are equally wrong in opposite directions, and beyond it the adjustment is worse.

Where $\text{Cov}(\varepsilon, z) = 0$, baseline imputation already recovers the covariance exactly and $\lambda$ is undefined. The adjustment preserves that recovery only if the predictor channel is also zero.

### 3.2.4 Multiple External Variables

Note that the variable $z$ is external to the first stage and only observed in the target sample. Because it takes no part in estimating $f$ or constructing $\eta$, one $\tilde{\eta}$ serves however many external variables the researcher has.

With $\mathbf{Z} = (z_1, \ldots, z_J)$, Equation (14) applies to each component,

$$\lambda_j = \frac{\text{Cov}(\tilde{\eta}, z_j)}{\text{Cov}(\varepsilon, z_j)}. \tag{25}$$

The numerator varies with how $z_j$ sorts across the predicted distribution, and the denominator with how much of its covariance with $y$ the first stage leaves out. Neither depends on the other components, so each covariance falls into its own regime. An estimand that combines several covariances is different. A regression coefficient on $z_j$, for instance, depends on the recovery fractions of all the components, as I show in Section 5.3.

# 4 Quantile Adjustment in Practice

The results of Section 3.2 concern population quantities. In practice the researcher observes a finite donor sample of $n_d$ observations and a finite target sample of $n_t$ observations, where $n_d$ and $n_t$ need not be equal.

Four practical steps separate the population results from these samples. The researcher has to estimate the quantile gap $\eta(p)$ from discrete empirical quantiles (Section 4.1), make sure target predictions fall within the support of the donor's predicted distribution (Section 4.2), deal with mass points in the predicted distribution (Section 4.3), and assess which regime the imputation is likely to fall in before applying the adjustment (Section 4.4).

## 4.1 Estimating the Quantile Gap

I estimate the first stage in the donor sample. In the applications that follow it takes the log-linear form,

$$\log y_d = \alpha + \mathbf{X}_d^\top \beta + \varepsilon_d, \tag{26}$$

with predictions $\hat{y}_g = \hat{\alpha} + \mathbf{X}_g^\top \hat{\beta}$ for $g \in \{d, t\}$. From here on $y$ denotes log income, so $\hat{y}_g$ is a prediction of log income and I compute the quantile gap on the log scale.

The empirical quantile gap compares the observed and predicted distributions in the donor sample,

$$\hat{\eta}(p) = \hat{Q}_{y_d}(p) - \hat{Q}_{\hat{y}_d}(p), \tag{27}$$

where $\hat{Q}$ denotes the sample quantile function.[3] I evaluate it on a grid of $G = 200$ positions $p \in \{p_1, \ldots, p_G\}$, uniformly spaced over the unit interval.[4]

The grid is discrete while target ranks are continuous, so I interpolate between grid points with a smoothing spline to obtain $\hat{\eta}(\cdot)$ at any $p \in [0, 1]$. Smoothing has a cost, since the adjusted values reproduce the marginal distribution of $y$ up to the smoothing error rather than exactly.

To apply the adjustment in the target sample, I give each target observation a quantile position relative to the donor's predicted distribution,

$$p_t = \hat{F}_{\hat{y}_d}(\hat{y}_t), \tag{28}$$

where $\hat{F}_{\hat{y}_d}(\cdot)$ is the empirical CDF of donor predictions. The adjusted outcome is

$$\tilde{y}_t = \hat{y}_t + \tilde{\eta}_t, \tag{29}$$

[3] As an alternative, one can fit a parametric distribution (such as the Singh–Maddala, Dagum or Pareto models) to $y_d$ and compute $\hat{\eta}(p)$ from the implied quantile functions, in the spirit of adjustments for missing top incomes (Alvaredo 2011; Jenkins 2017; Lustig 2019). This can help with tail corrections.

[4] Where the observed and predicted quantile functions are fairly smooth, a modest grid captures $\hat{\eta}(p)$. Where the tails are more irregular, a finer grid gives more accurate tail corrections; In Appendix D.1 I use $G = 2{,}000$ for income in pesos.

where $\tilde{\eta}_t := \hat{\eta}(p_t)$ is the realised adjustment at position $p_t$. Positions come from the donor's predicted distribution, the same distribution that enters $\hat{\eta}$, so a target observation receives the adjustment that a donor observation with the same predicted income would receive.

## 4.2 Common Support and Distributional Alignment

Each target observation gets its adjustment off the donor's quantile gap, so the quantile adjustment needs target predictions to lie within the support of the donor's predicted distribution. The common-population assumption of Section 2 delivers this in the population, but a finite donor sample need not cover the observed range of $\hat{y}_t$. When $\hat{y}_t$ falls outside that range, its quantile position sits at the boundary ($p_t \approx 0$ or $p_t \approx 1$), $\hat{\eta}(p_t)$ relies on extrapolation, and the adjustment for that observation may be unreliable.

The donor and target covariate distributions can also differ. In that case I suggest reweighting the donor sample before estimating $\hat{\eta}(p)$, so that its covariate distribution matches the target's. The weights $w_i$ satisfy

$$\sum_{i=1}^{n_d} w_i \cdot \mathbf{1}\{\mathbf{X}_i \leq \mathbf{x}\} \approx \hat{F}_{\mathbf{X}_t}(\mathbf{x}) \qquad \forall\, \mathbf{x}, \tag{30}$$

and can come from inverse probability weighting (DiNardo et al. 1996) or entropy balancing (Hainmueller 2012). They enter the donor's empirical quantiles,

$$\hat{\eta}(p) = \hat{Q}^w_{y_d}(p) - \hat{Q}^w_{\hat{y}_d}(p), \tag{31}$$

which anchors the adjustment to the target population's distribution. Estimating Equation (26) by weighted least squares with the same weights anchors the first-stage coefficients to that distribution as well.

## 4.3 Discrete Covariates and Bunching

Mass points in the distribution of $\hat{y}$ arise for two reasons. Some reflect population structure, and where individuals share the same covariate values it is appropriate to assign them the same $\tilde{\eta}$. Others are an artefact of a coarse prediction model, and these are the ones that cause trouble.

When $\mathbf{X}$ is discrete or low-dimensional, $\hat{f}(\mathbf{X})$ maps many distinct individuals to the same predicted value. Over the interval of $p$ that such a mass point occupies, $\hat{Q}_{\hat{y}}(p)$ is flat while $\hat{Q}_y(p)$ keeps rising, so $\hat{\eta}(p)$ varies across the interval. All tied observations nonetheless receive the same position $p_t$, and therefore the same adjustment, whatever their true $y$ or $z$. The adjusted values keep the mass point, and over that interval the adjusted distribution no longer reproduces the distribution of $y$.

The simplest fix is a richer prediction model. Adding covariates, interactions or flexible nonparametric methods increases the number of distinct predicted values and breaks mass points apart. Where the researcher cannot refine the covariate space further, I suggest inspecting $\hat{\eta}(p)$ directly and check that the curve is well-behaved.

## 4.4 Model Selection and Regime Diagnostics

I show in Section 3.2 that over-recovery becomes possible when little of the covariance between $y$ and $z_k$ remains for the adjustment to recover. A predictor can improve income prediction and at the same time explain a disproportionate share of $z_k$, which shrinks that remainder. Maximising first-stage $R^2$ is therefore not the objective, and the specification needs a check of its own.

### 4.4.1 Variable Selection

For predictor $i$ and external variable $z_k$, define the covariance-to-income contribution ratio

$$S_i(z_k) = \frac{R^2_{z_k,t}(\mathbf{X}) - R^2_{z_k,t}(\mathbf{X}_{-i})}{R^2_{y,d}(\mathbf{X}) - R^2_{y,d}(\mathbf{X}_{-i})}, \tag{32}$$

where $R^2_{y,d}(\cdot)$ is the first-stage $R^2$ in the donor sample and $R^2_{z_k,t}(\cdot)$ the $R^2$ from regressing $z_k$ on the same predictors in the target sample. The numerator is the loss in explanatory power for $z_k$ when predictor $i$ is left out, and the denominator the corresponding loss for $y$. A large $S_i(z_k)$ therefore identifies a predictor that contributes substantially to explaining $z_k$ while adding little to the explanation of $y$, the pattern that enlarges the predictor channel without improving covariance recovery.

Consider the intergenerational example again, and suppose the researcher adds offspring gender as a predictor. It could predict offspring income strongly through labour-market gender gaps while adding little to the prediction of parental income, since the offspring's gender has no bearing on a parent's own earnings history.

If any pair $(i, z_k)$ yields $S_i(z_k) > \tau$, I drop the predictor attaining $\max_{i,k} S_i(z_k)$, recompute all ratios on the reduced set, and repeat until no pair exceeds the cutoff $\tau$, so the screen works iteratively. My default is $\tau = 10$, which flags a predictor contributing an order of magnitude more to the explanation of $z_k$ than to that of $y$. A more conservative $\tau = 5$ removes more predictors. That lowers the risk of over-recovery, but it can discard predictors that would have improved $R^2_{y,d}$, and the coarser predictions that result can introduce mass points (Section 4.3).

Because a single $\mathbf{X}$ serves all external variables, I remove a predictor flagged for any $z_k$ from the specification for all of them, and I keep every predictor that survives the screen.

### 4.4.2 Assessing the Regime

To evaluate the regime, I use the value of the residual correlation $\rho = \mathrm{Corr}(\varepsilon, z_\perp)$ at which the adjustment would recover the omitted covariance exactly. By Equation (22), the denominator of $\lambda$ in Equation (18) equals $\rho\, \sigma_\varepsilon\, \sigma_{z_\perp}$, and since the error channel is zero its numerator equals $\mathrm{Cov}(\tilde{\eta}, z)$. Setting $\lambda = 1$ and solving for $\rho$ gives the threshold

$$\rho^* = \frac{\widehat{\mathrm{Cov}}(\tilde{\eta}_t, z_t)}{\hat{\sigma}_{\varepsilon,d}\, \hat{\sigma}_{z_\perp,t}}, \tag{33}$$

Each quantity comes from the sample that observes it. I estimate the numerator in the target sample, where both the adjustment and the external variable are available. The denominator combines the standard deviation of the first-stage residuals in the donor sample with that of the residuals from regressing $z_k$ on $\mathbf{X}$ in the target sample.[5] Dividing the two expressions gives $\lambda = \rho^*/\rho$.

The researcher never observes $\rho$, so she has to judge $\rho^*$ against theory and prior evidence on the dependence between $y$ and $z_k$ outside $\mathbf{X}$. The comparison gives a magnitude as well as a direction, with matching signs, a defensible $|\rho|$ above $|\rho^*|$ puts the application in partial covariance adjustment. One between $|\rho^*|/2$ and $|\rho^*|$ gives over-recovery that still improves on baseline imputation, and one at or below $|\rho^*|/2$ gives $\lambda \geq 2$, where the adjustment does no better than leaving the imputation alone. Opposite signs give mis-recovery.

I obtain intervals for $\rho^*$ by repeating the full variable selection and regime assessment across $B$ subsamples, drawn as I describe in Section 5.

## 4.5 Summary of the Algorithm

The complete procedure runs as follows. The `tstsqa` package in `R` at github.com/PedroToL/tstsqa implements each step.

1. If donor and target covariate distributions differ, reweight the donor sample (inverse probability weighting or entropy balancing) to match the target's.
2. Screen predictors with $S_i(z_k)$ (Equation (32)), refitting the first stage on the current predictor set at each round and removing one predictor at a time while any pair exceeds $\tau$. Keep all survivors.
3. Estimate $\widehat{f}(\cdot)$ in the donor sample on the surviving predictors.
4. Estimate the quantile gap $\widehat{\eta}(p)$ on the grid and smooth it with a spline.
5. Predict $\widehat{y}_t$ in the target sample, assign quantile positions $p_t$, and compute $\widetilde{\eta}_t = \widehat{\eta}(p_t)$.

[5] Where weights enter the quantiles, I use the same weights for these moments.

6. Compute $\rho^*$ (Equation (33)) and compare it with prior evidence on $\rho$. This step calls for judgement rather than a rule, and the researcher reports $\rho^*$ together with the judgement made from it.

7. Form the adjusted outcome $\tilde{y}_t = \hat{y}_t + \tilde{\eta}_t$.

# 5 Empirical Validation

Most empirical estimands are built from variances and covariances, so the two biases reach each estimand through its functional form. I evaluate the quantile adjustment in three applications that differ in the estimand and in the position imputed income takes in the second stage. The estimand is a correlation in Mexico (Section 5.1), an elasticity in Ecuador (Section 5.2), and the regression coefficients and inequality measures of an inequality of opportunity decomposition in Chile (Section 5.3).

Each application follows the same steps. I derive how the two biases enter the estimand, describe the data and report the benchmark, apply the screening and regime diagnostics of Section 4.4, and compare baseline imputation, stochastic augmentation and the quantile adjustment.

Comparing the three methods requires a setting in which the answer is known. I therefore construct the two-sample setting artificially, splitting a single dataset that observes $y$ and $z$ jointly. The full working sample plays the role of the population, and the full-sample estimate of each estimand serves as the benchmark. Because I draw donor and target from that same sample, a discrepancy between an imputed estimate and the benchmark reflects the imputation procedure and sampling variation across draws.[6]

Because the full sample observes $y$ and $z$ jointly, I can compute $\mathrm{Cov}(\varepsilon, z)$ and report the recovery fraction $\lambda$ alongside the results. A researcher facing a real two-sample problem cannot, which is why I develop $\rho^*$ in Section 4.4. I report both to show how the judgement

[6]Reading the results as informative about the population estimand requires $\hat{\theta} \xrightarrow{p} \theta$ under standard regularity conditions, which holds for each estimand considered.

based on $\rho^*$ compares with the realised regime where the comparison is possible.

The screen and the regime diagnostics run on the full working sample. For their intervals I draw $B = 100$ subsamples of $n^* = 30{,}000$ observations without replacement, use each one as both donor and target, and report the 2.5th and 97.5th percentiles of $S_i(z_k)$, $\rho^*$ and $\lambda$ across them. Each subsample plays both roles because $\rho$ and $\lambda$ require $y$ and $z$ for the same observations. I report the full screening output in Appendices D, E and F.

To compare the methods I use repeated sampling, in the spirit of a Monte Carlo analysis. At each of 1,000 replicates I draw $n_d + n_t$ observations from the full sample without replacement and split them into a donor sample of $n_d$ observations and a target sample of $n_t$, with no observation in both. I redraw the subsample at every replicate, so donor and target vary together across replicates instead of staying fixed while I resample within them. In each replicate I estimate the first stage on the donor, apply each method in the target, and compute each estimand.

I measure performance by the bias ratio,

$$\mathrm{BR}_a(\theta) = \frac{\theta(\tilde{y}_a, z)}{\theta(y, z)}, \tag{34}$$

where $a \in \{\text{baseline}, \text{stochastic}, \text{quantile}\}$. A value of one indicates alignment with the benchmark, values below one understatement and values above one overstatement. For each method I report the mean bias ratio across replicates and the 2.5th and 97.5th percentiles of its distribution.

The spread across replicates therefore measures how much each method's deviation from the benchmark moves when the donor and target are redrawn from the same working sample. In Section 6.1 I summarise these as relative bias and relative RMSE across the applications. It is not a standard error for the estimand. It excludes the sampling uncertainty of the benchmark, and it says nothing about what would happen were donor and target drawn from different populations, or were either measured with error.

## 5.1 Joint Distributions

The first application concerns the joint distribution of income and consumption in Mexico, an estimand where both biases operate at once. Researchers often summarise the joint distribution of income and another economic outcome by the Pearson correlation coefficient (Decancq 2014; Brewer et al. 2017; Fasianos and Lydon 2021; Donatiello et al. 2025). With imputed income $\tilde{y}$ in place of $y$, the bias ratio is

$$\frac{\rho_{\text{TSTS}}}{\rho_{\text{benchmark}}} = \frac{\text{Cov}(\tilde{y}, z)}{\text{Cov}(y, z)} \cdot \frac{\sigma_y}{\sigma_{\tilde{y}}}. \tag{35}$$

The first factor carries the covariance bias and the second the variance bias. When $\text{Cov}(\varepsilon, z) > 0$, covariance bias pulls the first factor below one. Variance bias pushes the second above one, since a compressed $\tilde{y}$ has $\sigma_{\tilde{y}} < \sigma_y$.

For methods that restore the variance of $y$, the second factor equals one and the bias ratio reduces to

$$\frac{\rho_{\text{TSTS}}}{\rho_{\text{benchmark}}} = \frac{\text{Cov}(\hat{y}, z) + \lambda\ \text{Cov}(\varepsilon, z)}{\text{Cov}(y, z)}. \tag{36}$$

Stochastic augmentation restores the variance but recovers none of the omitted covariance, which corresponds to $\lambda = 0$ in Equation (36) (Section 3.1), so its bias ratio lies below one whenever $\text{Cov}(\varepsilon, z) > 0$. The quantile adjustment restores the variance and recovers a fraction $\lambda$ of the omitted covariance. Baseline imputation corrects neither bias, and its two factors in Equation (35) pull in opposite directions.

### 5.1.1 Data

I use the 2024 National Survey of Household Income and Expenditure (ENIGH), which records both income and consumption expenditure for Mexican households. After trimming income at the top and bottom 0.05 per cent and dropping non-positive values of income and expenditure, the working sample contains $n = 91{,}242$ households, representing a total of 38,771,194 households.

In each replicate I draw 50,000 households at random from the working sample and split them into a donor sample ($n_d = 30{,}000$) with observed income $y$ and a target sample ($n_t = 20{,}000$) with observed consumption $z$. Both samples share nine candidate predictors $\mathbf{X}$: household head education and gender, locality size, dwelling type, household size, share of female members, share of occupied members, share of members receiving income, and state fixed effects. I summarise the working sample in Appendix Table D.1.

The benchmark is a survey estimate, so every moment, quantile function and regression in the exercise uses the survey's sampling weights, in the benchmark and in each replicate alike. Donor and target come from the same survey, so their covariate distributions differ only by sampling variation, and I do not need the reweighting of Section 4.2.

I estimate the first stage in log-linear form, and both income and consumption enter the correlation in logs. In Appendix D.1 I report the same exercise with the estimands defined in levels.

On the full working sample, which observes income and consumption jointly, the benchmark moments are $\mathrm{Var}(\log y) = 0.525$, $\mathrm{Cov}(\log y, \log z) = 0.395$, and

$$\rho_{\text{benchmark}} = \frac{\mathrm{Cov}(\log y, \log z)}{\sigma_{\log y}\,\sigma_{\log z}} = 0.737. \tag{37}$$

### 5.1.2 Variable Selection and Regime Diagnostics

I screen the nine candidate predictors against log consumption. None exceeds the cutoff $\tau = 10$, so I retain all of them. The largest mean ratio across subsamples is for the share of female members, at $S = 7.87$, followed by household head gender at 1.42 (Appendix Table D.2). The share of female members also has the widest interval, [1.03, 37.93], and its upper end lies above $\tau$.

With these predictors the first stage attains $R^2 = 0.479$, and the threshold is $\rho^* = 0.341$. The question is whether the dependence between income and consumption that household characteristics leave unexplained is stronger than that. What household characteristics fail

to explain about income is more likely to be persistent than transitory, and consumption tracks permanent income rather than these characteristics alone (Friedman 1957; Hall 1978; Blundell et al. 2008). The persistent component the first stage misses in predicting income therefore reappears in consumption, which leads me to expect a residual correlation above 0.341.

The split design lets me check this judgement. The realised residual correlation is $\rho = 0.557$, above $\rho^*$ and of the same sign, so the application falls in partial covariance adjustment, with $\lambda = 0.612$ [0.595, 0.630] across subsamples. Drawing donor and target independently, as a researcher would, gives $\rho^* = 0.340$ [0.330, 0.350], close to the full-sample value. I collect these quantities with their subsample intervals in Appendix Table D.3.

### 5.1.3 Results

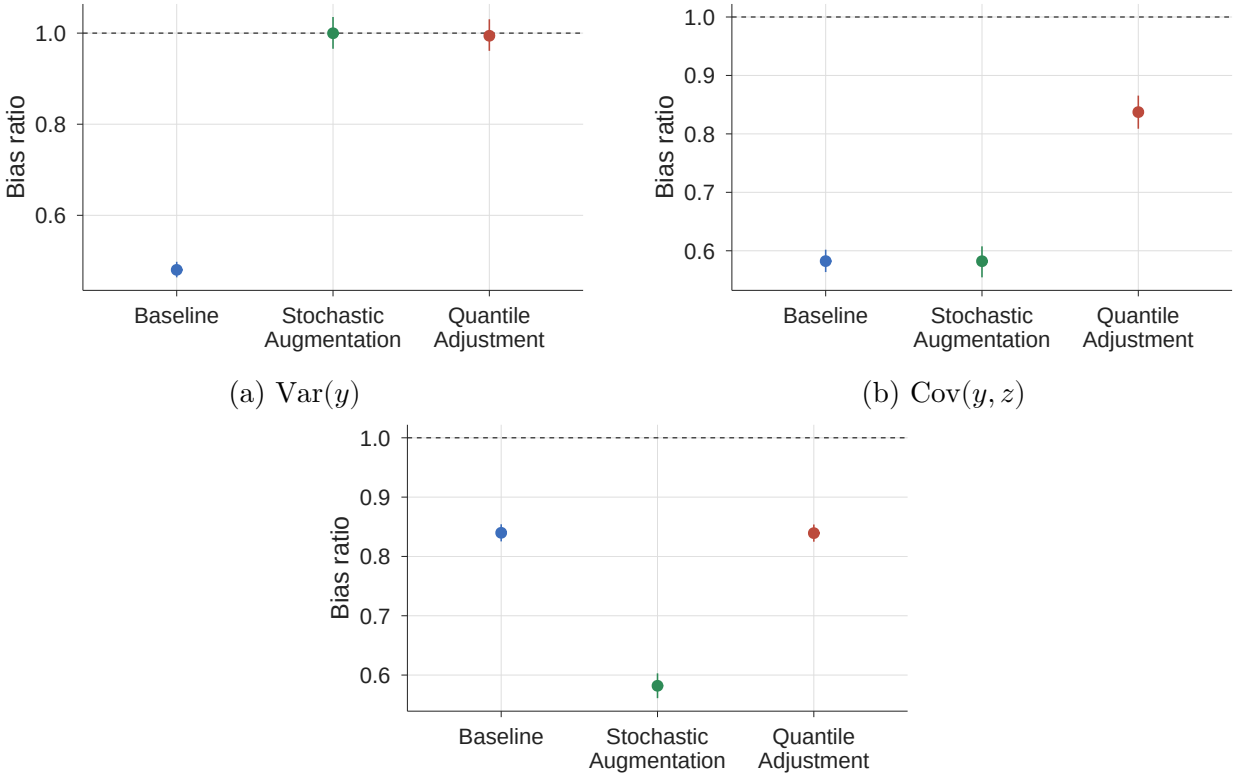
I report the point estimates in Table 2 and the bias ratios in Figure 4. In Appendix Figures D.1 and D.2 I show the estimated quantile gap and the resulting income densities for a single replicate.

Table 2: Mexico: Income–Consumption Estimates

| | Benchmark | Baseline | Stochastic | Quantile |
|---|---|---|---|---|
| $\mathrm{Var}(y)$ | 0.525 | 0.252 [0.244, 0.261] | 0.525 [0.507, 0.544] | 0.522 [0.505, 0.541] |
| $\mathrm{Cov}(y, z)$ | 0.395 | 0.230 [0.223, 0.238] | 0.230 [0.219, 0.240] | 0.331 [0.319, 0.342] |
| $\rho_{y,z}$ | 0.737 | 0.619 [0.608, 0.630] | 0.429 [0.414, 0.444] | 0.619 [0.608, 0.629] |
| $R^2$ | | 0.480 [0.470, 0.490] | | |
| $\lambda$ | | 0.612 [0.582, 0.643] | | |

*Notes:* Benchmark values come from the full working sample. Brackets give the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. $R^2$ and $\lambda$ are computed within each replicate. Income and consumption in logs.

**Variance and covariance.** Baseline imputation compresses the variance to 0.252, the benchmark variance multiplied by the first-stage $R^2$ ($0.525 \times 0.480$), and recovers 58 per




(a) Var($y$)

(b) Cov($y, z$)

(c) $\rho_{y,z}$

Figure 4: Mexico: Bias Ratios

*Notes:* Bias ratio for each moment under the three imputation methods. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. Income and consumption in logs.

cent of the covariance. Stochastic augmentation restores the benchmark variance and leaves the covariance where baseline imputation put it, as Section 3.1 predicts.

The quantile adjustment restores the variance and recovers 84 per cent of the covariance, about 26 percentage points more than the other two methods. The recovery fraction accounts for the difference. Baseline imputation leaves 42 per cent of the covariance unrecovered, and the adjustment puts back $\lambda = 0.612$ of that remainder, so $0.582 + 0.612 \times 0.418 = 0.838$. The two percentages answer different questions. The recovery fraction $\lambda$ is the share of the *missing* covariance the adjustment recovers, and 84 per cent the share of the *total* covariance the adjusted estimate reaches.

**Correlation coefficient.** The two biases enter the correlation in opposite directions. Under baseline imputation the covariance is 42 per cent short, while the standard deviation of imputed income is 0.693 of the benchmark ($\sqrt{0.480}$), which raises the ratio of standard deviations in Equation (35) by 44 per cent. The two nearly offset, and the correlation ends 16 per cent below the benchmark ($0.582 \times 1.443 = 0.840$).

The quantile adjustment removes almost all of the variance bias and part of the covariance bias, and also ends 16 per cent below the benchmark. The two methods land a tenth of a percentage point apart through different mechanisms, and I examine why in Section 6.2.

Stochastic augmentation removes the variance bias without touching the covariance. That removes the offsetting term and exposes the full covariance attenuation, so its correlation bias equals its covariance bias, at 42 per cent, and it does worse than leaving the imputation unadjusted.

## 5.2 Elasticities

The preceding application treated $y$ and $z$ symmetrically. Here imputed income sits on the right-hand side of the second stage, as the regressor whose variance enters the estimand directly. The intergenerational elasticity regresses offspring income on parental income (Solon 1992; Björklund and Jäntti 1997),

$$\beta_{\text{IGE}} = \frac{\text{Cov}(y, z)}{\text{Var}(y)}, \tag{38}$$

where $y$ is parental income and $z$ offspring income. Parental income is the imputed variable, so both biases enter and work against each other. Covariance attenuation shrinks the numerator, while variance compression shrinks the denominator.

Which bias dominates decides the direction of the error. Writing $s$ for the share of the covariance the first stage omits, baseline imputation gives

$$\frac{\beta_{\text{baseline}}}{\beta_{\text{benchmark}}} = \frac{1 - s}{R^2}, \tag{39}$$

which exceeds one whenever $R^2 < 1 - s$. Stochastic augmentation restores the denominator and leaves the numerator, so its ratio is $1 - s$ and it understates whenever $s > 0$. The quantile adjustment restores the denominator and recovers a share $\lambda$ of what the numerator lost, giving $1 - s(1 - \lambda)$.

### 5.2.1 Data

I use Ecuadorian administrative records linked through a pseudonymised personal identifier. Monthly social security records from the Laboratory of Labour and Business Dynamics (LDLE) cover 2009 to 2023 and supply wages, the civil registry links offspring to their parents, and the 2022 population and housing census supplies education, language and province of residence. The sample therefore covers families in which the offspring and at least one parent appear in the social security records.

I measure permanent income as the average monthly wage across all the years a person is observed between ages 15 and 65, where each year's wage is a months-weighted average across employers. Parental income is the average of the father's and the mother's permanent income where both are observed, and the observed parent's income otherwise. After trimming parental income at the top and bottom 0.05 per cent and dropping pairs with non-positive income on either side, the working sample contains $n = 281{,}468$ parent–offspring pairs.

In each replicate I draw 60,000 pairs at random from the working sample and split them into a donor sample ($n_d = 30{,}000$) with observed parental income $y$ and a target sample ($n_t = 30{,}000$) with observed offspring income $z$. Both samples share six candidate predictors $\mathbf{X}$, all measured on the parents: age, gender, years of education, indigenous language, employment sector and natural region of residence.

Where both parents are observed, age and years of education are averaged, the language indicator equals one if either parent speaks an indigenous language, and the categorical predictors take the father's value. I summarise the working sample in Appendix Table E.5.

The registers carry no sampling weights, so every moment, quantile function and regression

in this application is unweighted. As in the other applications, donor and target come from the same source, and I do not need the reweighting of Section 4.2. I estimate the first stage in log-linear form, and both incomes enter the elasticity in logs.[7]

On the full working sample, which observes both incomes jointly, the benchmark values are $\mathrm{Var}(\log y) = 0.644$, $\mathrm{Cov}(\log y, \log z) = 0.258$, a correlation of 0.415, and

$$\beta_{\text{benchmark}} = \frac{\mathrm{Cov}(\log y, \log z)}{\mathrm{Var}(\log y)} = 0.401. \tag{40}$$

### 5.2.2 Variable Selection and Regime Diagnostics

I screen the six candidate predictors against log offspring income. None exceeds the cutoff $\tau = 10$, so I retain all of them. The largest mean ratio across subsamples is for indigenous language, at $S = 2.90$, followed by age at 1.89, and every 97.5th percentile lies below the cutoff (Appendix Table E.6).

With these predictors the first stage attains $R^2 = 0.648$, the highest of the three applications, and the threshold is $\rho^* = 0.152$. The question is whether the dependence between parental and offspring income that parental characteristics leave unexplained is stronger than that.

Parental income here is a permanent measure, so the transitory variation that attenuates elasticities estimated on a single year is largely averaged out (Mazumder 2005), and what the predictors miss is mostly a persistent difference in standing. Channels such as wealth and networks carry that standing to the next generation and lie outside the predictors, which leads me to expect a residual correlation above 0.152.

The split design lets me check this judgement. The realised residual correlation is $\rho = 0.175$, above $\rho^*$ and of the same sign, so the application falls in partial covariance adjustment, with $\lambda = 0.868$ [0.815, 0.938] across subsamples. Drawing donor and target independently gives $\rho^* = 0.152$ [0.144, 0.159], the same as the full-sample value. I collect these quantities with

[7] The selectivity ratios reported here and in Appendix Table E.6 are computed on simulated data rather than on the administrative records, so the screening is provisional. Every other estimate in this section comes from the records.

their subsample intervals in Appendix Table E.7.

### 5.2.3 Results

I report the point estimates in Table 3 and the bias ratios in Figure 5. In Appendix Figures E.6 and E.7 I show the estimated quantile gap and the resulting income densities for a single replicate.

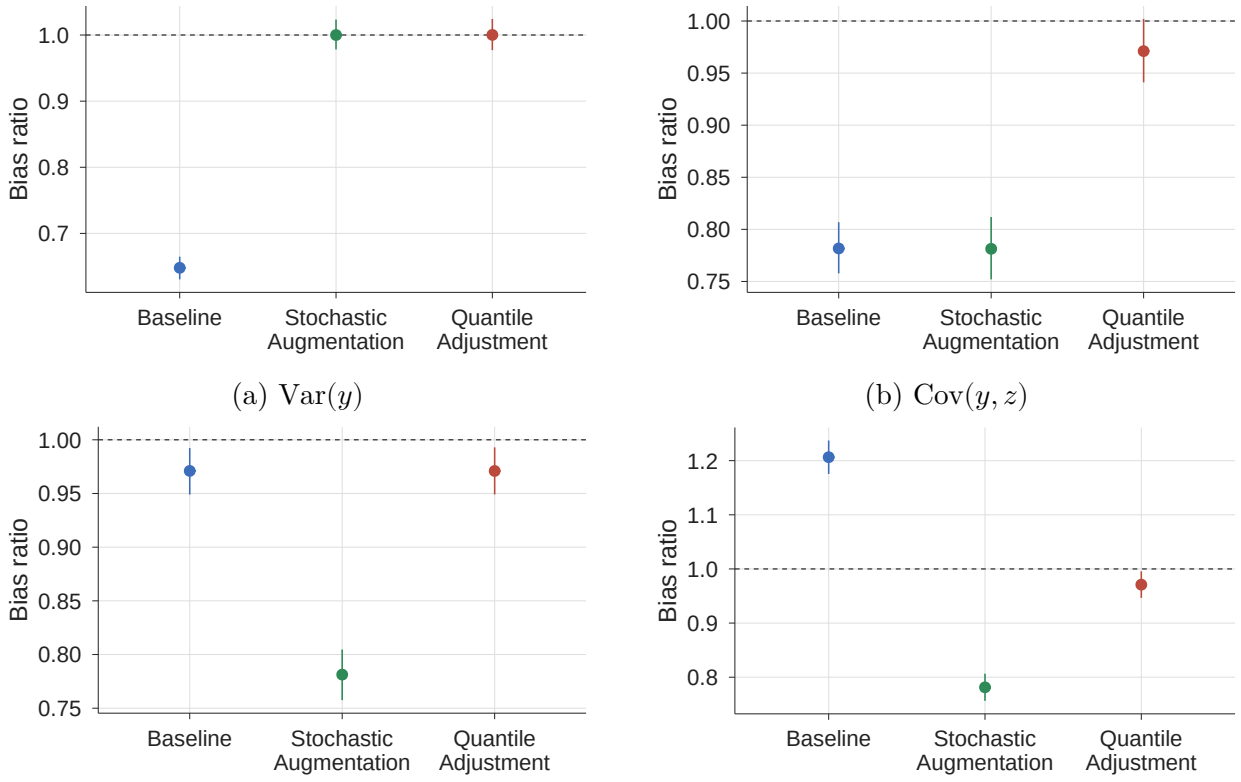
Table 3: Ecuador: Intergenerational Elasticity Estimates

| | Benchmark | Baseline | Stochastic | Quantile |
|---|---|---|---|---|
| $\mathrm{Var}(y)$ | 0.644 | 0.417<br>[0.406, 0.428] | 0.644<br>[0.630, 0.659] | 0.644<br>[0.629, 0.660] |
| $\mathrm{Cov}(y, z)$ | 0.258 | 0.202<br>[0.196, 0.208] | 0.202<br>[0.194, 0.210] | 0.251<br>[0.243, 0.259] |
| $\rho_{y,z}$ | 0.415 | 0.403<br>[0.394, 0.412] | 0.324<br>[0.314, 0.334] | 0.403<br>[0.394, 0.412] |
| $\beta_{\mathrm{IGE}}$ | 0.401 | 0.484<br>[0.471, 0.496] | 0.313<br>[0.303, 0.323] | 0.389<br>[0.379, 0.399] |
| $R^2$ | | 0.648 [0.641, 0.654] | | |
| $\lambda$ | | 0.867 [0.804, 0.940] | | |

*Notes:* Benchmark values come from the full working sample. Brackets give the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. $R^2$ and $\lambda$ are computed within each replicate. Both incomes in logs.

**Variance and covariance.** Baseline imputation compresses the variance to 0.417, the benchmark variance multiplied by the first-stage $R^2$ ($0.644 \times 0.648$), and recovers 78 per cent of the covariance. Stochastic augmentation restores the variance and leaves the covariance where baseline imputation put it. The quantile adjustment restores the variance and recovers 97 per cent of the covariance. The first stage omits 22 per cent of it, and the adjustment puts back $\lambda = 0.868$ of that remainder, so $0.782 + 0.868 \times 0.218 = 0.971$.

**Elasticity.** Baseline imputation overstates the elasticity by 20.6 per cent. Its covariance shortfall of 21.8 per cent is smaller than its variance shortfall of 35.2 per cent, and Equation (39) gives $0.782/0.648 = 1.206$. An elasticity that overstates persistence understates




(a) $\mathrm{Var}(y)$ (b) $\mathrm{Cov}(y, z)$

(c) $\rho_{y,z}$ (d) $\beta_{\mathrm{IGE}}$

Figure 5: Ecuador: Bias Ratios

*Notes:* Bias ratio for each estimand under the three imputation methods. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. Both incomes in logs.

mobility.

Stochastic augmentation removes the variance compression and leaves the covariance attenuation, so it understates the elasticity by 21.9 per cent, an error slightly larger than that of baseline imputation. The quantile adjustment lands 2.9 per cent below the benchmark, as $1 - 0.218 \times (1 - 0.868) = 0.971$ implies.

**Correlation.** Baseline imputation comes within 2.9 per cent of the benchmark correlation, because its two biases nearly offset ($0.782/\sqrt{0.648} = 0.971$). Stochastic augmentation removes the offset and ends 21.9 per cent below. The quantile adjustment also ends 2.9 per cent below, within a hundredth of a percentage point of baseline imputation. I explain

this agreement in Section 6.2. It arises here even though the quantile gap is not monotone (Appendix E).

In Appendix E.1 I report the rank-rank slope as an alternative specification.

## 5.3 Dependent Variable and Inequality Measures

The two preceding applications place $y$ symmetrically with $z$, as in a correlation, and on the right-hand side of the second stage, as in an elasticity. The third places it on the left-hand side, as the dependent variable, which completes the positions imputed income can take in a second stage.

I use inequality of opportunity (IOp) for this case for two reasons. The IOp regression puts income on a vector of circumstances rather than on a single external variable, which exercises the extension of Section 3.2.4. And IOp studies report inequality measures alongside the regression coefficients, so the exercise includes estimands that depend on the tails and shape of the distribution as well as its variance.

IOp splits total income inequality into a component attributable to circumstances beyond individual control, such as parental education, ethnicity or region of birth, and a residual component (Roemer and Trannoy 2016). In the parametric ex ante version, the researcher regresses income on a vector of circumstances $\mathbf{Z}$ and builds the opportunity-determined component from the fitted values.

The application has four estimands. They are the circumstance coefficients $\delta$, total inequality $I(y)$, absolute IOp $I(y_{|\mathbf{z}})$, and the relative share $\theta = I(y_{|\mathbf{z}})/I(y)$. Because $\mathbf{Z}$ is a vector, each circumstance $z_j$ has its own recovery fraction $\lambda_j$ (Equation (25)).

### Circumstance Coefficients

Because the target sample observes $\mathbf{Z}$, imputation leaves $\mathrm{Var}(\mathbf{Z})$ unaffected. In the scalar case, $\delta_{\text{benchmark}} = \mathrm{Cov}(y, z)/\mathrm{Var}(z)$ depends on the imputation only through the covariance,

so variance bias drops out entirely and the bias ratio is

$$\frac{\delta_{\text{TSTS}}}{\delta_{\text{benchmark}}} = \frac{\text{Cov}(\hat{y}, z) + \lambda \text{ Cov}(\varepsilon, z)}{\text{Cov}(y, z)}. \tag{41}$$

Baseline imputation and stochastic augmentation both leave the covariance at $\text{Cov}(\hat{y}, z)$, the $\lambda = 0$ case, so they return the same coefficient, and only the quantile adjustment shifts it.

With several circumstances, each coefficient combines the covariances of income with all of them. Writing $\delta = \text{Var}(\mathbf{Z})^{-1} \text{Cov}(\mathbf{Z}, y)$ and taking the $j$th row,

$$\delta_j = \sum_k w_{jk} \text{ Cov}(y, z_k), \tag{42}$$

where $w_{jk}$ is the $(j, k)$ element of $\text{Var}(\mathbf{Z})^{-1}$. When the circumstances are mutually uncorrelated the off-diagonal weights are zero, and $\delta_j$ depends on $\text{Cov}(y, z_j)$ alone. Otherwise they are nonzero and net out the part of that covariance the other circumstances already account for.

Substituting the imputed covariances gives the bias ratio

$$\frac{\delta_{j,\text{TSTS}}}{\delta_{j,\text{benchmark}}} = \frac{\sum_k w_{jk} \left[\text{Cov}(\hat{y}, z_k) + \lambda_k \text{ Cov}(\varepsilon, z_k)\right]}{\sum_k w_{jk} \text{Cov}(y, z_k)}. \tag{43}$$

Every recovery fraction enters every coefficient, so even with $\lambda_j = 1$ the coefficient on $z_j$ inherits the errors in the other circumstances' covariances. Correcting one coefficient at a time therefore does not suffice, because the coefficients are right only if the whole set of covariances is. The quantile adjustment acts on all of them at once, because it changes the imputed values rather than the estimand, and how close each covariance comes to its benchmark depends on its own $\lambda_k$.

### Total Inequality

Total inequality is the first of these estimands for which the level scale matters. I apply Duan's smearing correction to baseline imputation. Stochastic augmentation performs it

implicitly, as I note in Section 3.1, and the quantile adjustment needs none.

When the log errors are homoskedastic, the smearing retransformation makes $\mathbb{E}[y \mid \hat{y}] = \hat{y}$ hold on the level scale, so the distribution of true income is a mean-preserving spread of the distribution of imputed income.[8] The vector of realised incomes then majorizes the vector of imputed values, and since all standard inequality indices that satisfy the Pigou–Dalton transfer principle are Schur-convex (Marshall et al. 2011),

$$I(\hat{y}) \leq I(y), \tag{44}$$

with equality only where $\hat{y} = y$.

Stochastic augmentation restores the variance in logs but not the higher moments, so the log distribution it produces need not match the observed one in shape. The restoration also does not survive the move to levels, because exponentiation is nonlinear and the variance of $\exp(\tilde{y}_s)$ depends on moments the method does not control.[9] The quantile adjustment reproduces the distribution by construction, on either scale.

### Absolute and Relative IOp

Absolute IOp is the inequality of the predicted component, $I(\delta^\top \mathbf{Z})$, which asks how unequal incomes would be if circumstances were the only thing that varied. Because IOp studies commonly estimate the regression in logs (Ferreira and Gignoux 2011), I exponentiate the fitted values before applying the index.[10]

The absolute measure therefore inherits whatever the coefficients inherit. Where the $\lambda_j$ leave the coefficients attenuated, the predicted component is compressed and $I(\tilde{y}_{|\mathbf{z}}) < I(y_{|\mathbf{z}})$. Variance bias reaches this measure only through the coefficients, and they carry none, so

[8] In Appendix D.1 I discuss the heteroskedastic case, where a single smearing constant restores the mean but not the conditional mean.

[9] Under lognormality the Gini has the closed form $G = 2\Phi(\sigma/\sqrt{2}) - 1$, where $\sigma^2$ is the variance of log income, and restoring $\sigma^2$ suffices to recover the Gini. Outside lognormality, the Gini and other measures depend on higher moments and tail properties.

[10] This step needs no further correction, since smearing multiplies every fitted value by a constant and the Gini is scale-invariant.

baseline imputation and stochastic augmentation return the same value here as well.

The relative measure normalises by total inequality and so combines both biases,

$$\frac{\theta_{\text{TSTS}}}{\theta_{\text{benchmark}}} = \frac{I(\tilde{y}_{|\mathbf{z}})}{I(y_{|\mathbf{z}})} \cdot \frac{I(y)}{I(\tilde{y})}. \tag{45}$$

The numerator carries the coefficient attenuation and the denominator the compression of the income distribution, so the two work in opposite directions. Baseline imputation compresses both, and the errors partly cancel. Stochastic augmentation restores $I(y)$ without touching the covariances, which removes the cancellation and pushes the relative share below the baseline value. The quantile adjustment addresses both, and the $\lambda_j$ govern how much it recovers.

### 5.3.1 Data

I use the 2015 National Socioeconomic Characterization Survey (CASEN), which records household income alongside the four circumstances. After trimming income at the top and bottom 0.05 per cent and dropping non-positive values, the working sample contains $n =$ 98,950 observations, representing a weighted total of 6,704,013.

In each replicate I draw 70,000 observations at random from the working sample and split them into a donor sample ($n_d = 30{,}000$) with observed income $y$ and a target sample ($n_t = 40{,}000$) with observed circumstances $\mathbf{Z}$. Both samples share six candidate predictors $\mathbf{X}$: household size, share of male members, share of occupied members, household head education and gender, and regional fixed effects. The circumstance vector comprises four binary variables: the respondent's sex, indigenous status, and indicators for the father and for the mother having completed secondary education or above. I summarise the working sample in Appendix Table F.9.

As in the Mexican application, the benchmark is a survey estimate, so every moment, quantile function and regression uses the survey's sampling weights, in the benchmark and in each replicate alike, and I do not need the reweighting of Section 4.2. I estimate the first stage in

log-linear form, run the circumstance regression in logs, and measure inequality throughout with the Gini coefficient.

On the full working sample, which observes income and circumstances jointly, the benchmark values are $\mathrm{Var}(\log y) = 0.678$, total inequality $I(y) = 0.482$, absolute IOp $I(y_{|\mathbf{z}}) = 0.161$, and

$$\theta_{\text{benchmark}} = \frac{I(y_{|\mathbf{z}})}{I(y)} = 0.334. \tag{46}$$

I report the benchmark covariances and circumstance coefficients in Table 5.

### 5.3.2 Variable Selection and Regime Diagnostics

The screen removes two of the six candidate predictors, both flagged against the respondent's sex (Appendix Table F.10). The share of male household members goes first, with a mean ratio across subsamples of $S = 6.9 \times 10^7$, and household head gender second, at $S = 54.6$. The first ratio is this large because the share of male members adds almost nothing to the first-stage $R^2$, so the denominator of Equation (32) is close to zero.

Once the screen removes these two, the largest remaining ratio is region against indigenous status, at 2.38, and household size, share of occupied members, household head education and region survive (Appendix Table F.11). I return to what the screen does here in Section 6.3.

With those predictors the first stage attains $R^2 = 0.481$. Each circumstance has its own threshold, and in magnitude the four range from 0.034 to 0.141 (Table 4). Each circumstance is an attribute of the respondent while income is pooled at the household level, so in each case the question is how much of the attribute's relationship with income survives once household composition is in the specification.

For the two parental education indicators I expect a good deal to survive. The income the predictors fail to explain reflects the schooling, occupation and networks a household inherits, and parental education proxies for those. That reasoning implies residual correlations above the thresholds of 0.141 and 0.130.

For indigenous status the threshold is the smallest in magnitude, at $-0.034$. Region is the strongest predictor of ethnicity available and it is in the specification, yet indigenous individuals differ from others within a region in ways the predictors do not capture. That within-region variation survives into the residual, and I expect it to produce a residual correlation larger in magnitude than 0.034.

The respondent's sex is the uncertain case. Household size and the share of occupied members already absorb most of what sex says about a pooled income, and what remains runs through structures those predictors do not resolve, such as single-adult households headed by women. That channel is narrow, and whether it produces a residual correlation above 0.056 is not clear either way.

The split design lets me check these judgements, and I report the realised residual correlations alongside the thresholds in Table 4. The three circumstances I expected to fall in partial covariance adjustment do so, with $\lambda$ between 0.539 and 0.986. The respondent's sex, which I left open, over-recovers at $\lambda = 1.282$. Mother's education sits closest to the boundary, with a threshold 1.4 per cent below the realised correlation.

For these two circumstances the subsample intervals in Appendix Table F.12 are wide. The interval for mother's education, [0.819, 1.222], crosses one, and the interval for the respondent's sex, [0.859, 2.218], reaches from partial covariance adjustment past $\lambda = 2$.

Table 4: Chile: Regime Diagnostics by Circumstance

| Circumstance | $\rho^*$ | $\rho$ | $\lambda$ | Regime |
|---|---|---|---|---|
| Respondent male | 0.056 | 0.044 | 1.282 | Over-recovery |
| Indigenous | $-0.034$ | $-0.064$ | 0.539 | Partial covariance adjustment |
| Father secondary or more | 0.141 | 0.172 | 0.819 | Partial covariance adjustment |
| Mother secondary or more | 0.130 | 0.132 | 0.986 | Partial covariance adjustment |

*Notes:* $\rho^*$ is the threshold from Equation (33) and $\rho$ the realised residual correlation, both computed on the full working sample. $\lambda = \rho^*/\rho$ is the share of the missing covariance the adjustment recovers. The regime follows from comparing the two at matching signs.

### 5.3.3 Results

I report the circumstance coefficients, the inequality measures and the relative share in Table 5, and the corresponding bias ratios in Figures 6 and 7. In Appendix F I report the variance and covariances behind the coefficients, along with the quantile gap and income densities for a single replicate.

**Variance and covariances.** Panel A repeats the pattern of the Mexican application. Baseline imputation compresses the variance and partly recovers each covariance, stochastic augmentation restores the variance and leaves the covariances where baseline imputation put them, and the quantile adjustment improves on baseline imputation for all five moments. Only the covariance with the respondent's sex crosses the benchmark, ending 7.7 per cent above it. I report the corresponding bias ratios in Appendix Figure F.9.

For the respondent's sex, baseline imputation leaves the covariance 26.9 per cent short of the benchmark, so the first stage omits about 27 per cent of it. A recovery fraction above one overshoots only that omitted part. With the full-sample recovery fraction of 1.282 (Table 4), the adjusted covariance lands about $0.282 \times 26.9 \approx 7.6$ per cent above the benchmark, close to the replicate average of 7.7 per cent. The adjustment stops improving on baseline imputation at $\lambda = 2$. The replicate mean of $\lambda$ for this circumstance is 1.340, and its 97.5th percentile, 2.080, lies above that point.

**Circumstance coefficients.** Baseline imputation and stochastic augmentation return the same four coefficients, attenuated by between 26 and 44 per cent. The quantile adjustment moves all four toward the benchmark and cuts the largest error, for indigenous status, from 44 to 24 per cent. The coefficients on the respondent's sex and mother's education end above the benchmark, by 8.9 and 12.4 per cent. For both, the replicate interval contains the benchmark, so at these sample sizes a single application could land on either side of it. The other two coefficients remain short of the benchmark by margins their intervals exclude.

For the respondent's sex the overshoot follows from its own covariance, which the adjustment over-recovers ($\lambda = 1.340$). Mother's education recovers its own covariance almost exactly, at

Table 5: Chile: Inequality of Opportunity Estimates

| | Benchmark | Baseline | Stochastic | Quantile |
|---|---|---|---|---|
| *Panel A: Variance and covariances* | | | | |
| $\text{Var}(y)$ | 0.678 | 0.326 | 0.678 | 0.677 |
| | | [0.312, 0.341] | [0.653, 0.705] | [0.637, 0.719] |
| $\text{Cov}(y, \text{Male})$ | 0.048 | 0.035 | 0.035 | 0.051 |
| | | [0.030, 0.040] | [0.028, 0.042] | [0.043, 0.060] |
| $\text{Cov}(y, \text{Indigenous})$ | −0.023 | −0.013 | −0.013 | −0.019 |
| | | [−0.015, −0.012] | [−0.016, −0.010] | [−0.021, −0.016] |
| $\text{Cov}(y, \text{Father})$ | 0.081 | 0.051 | 0.051 | 0.075 |
| | | [0.046, 0.056] | [0.045, 0.057] | [0.068, 0.084] |
| $\text{Cov}(y, \text{Mother})$ | 0.061 | 0.040 | 0.040 | 0.060 |
| | | [0.036, 0.045] | [0.035, 0.047] | [0.053, 0.069] |
| *Panel B: Circumstance coefficients* | | | | |
| $\delta_{\text{Male}}$ | 0.177 | 0.131 | 0.131 | 0.193 |
| | | [0.113, 0.149] | [0.105, 0.157] | [0.165, 0.222] |
| $\delta_{\text{Indigenous}}$ | −0.257 | −0.144 | −0.143 | −0.196 |
| | | [−0.168, −0.119] | [−0.182, −0.103] | [−0.231, −0.161] |
| $\delta_{\text{Father}}$ | 0.610 | 0.363 | 0.364 | 0.533 |
| | | [0.329, 0.397] | [0.310, 0.422] | [0.474, 0.590] |
| $\delta_{\text{Mother}}$ | 0.396 | 0.291 | 0.290 | 0.445 |
| | | [0.250, 0.331] | [0.221, 0.359] | [0.376, 0.518] |
| *Panel C: Inequality measures* | | | | |
| $I(y)$ | 0.482 | 0.320 | 0.451 | 0.478 |
| | | [0.313, 0.327] | [0.440, 0.465] | [0.459, 0.498] |
| $I(y\|\mathbf{z})$ | 0.161 | 0.098 | 0.098 | 0.154 |
| | | [0.091, 0.106] | [0.088, 0.109] | [0.140, 0.171] |
| $\theta$ | 0.334 | 0.307 | 0.218 | 0.323 |
| | | [0.287, 0.329] | [0.197, 0.240] | [0.297, 0.351] |
| $R^2$ | | 0.481 [0.468, 0.494] | | |
| $\lambda_{\text{Male}}$ | | 1.340 [0.884, 2.080] | | |
| $\lambda_{\text{Indigenous}}$ | | 0.543 [0.417, 0.692] | | |
| $\lambda_{\text{Father}}$ | | 0.828 [0.663, 1.023] | | |
| $\lambda_{\text{Mother}}$ | | 1.002 [0.759, 1.329] | | |

*Notes:* Benchmark values come from the full working sample. Brackets give the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. Panels A and B use the log scale, and Panel C the level scale, with the Gini coefficient as the inequality measure. $R^2$ and each $\lambda_j$ are computed within each replicate; I report their full-sample counterparts in Appendix Table F.12.

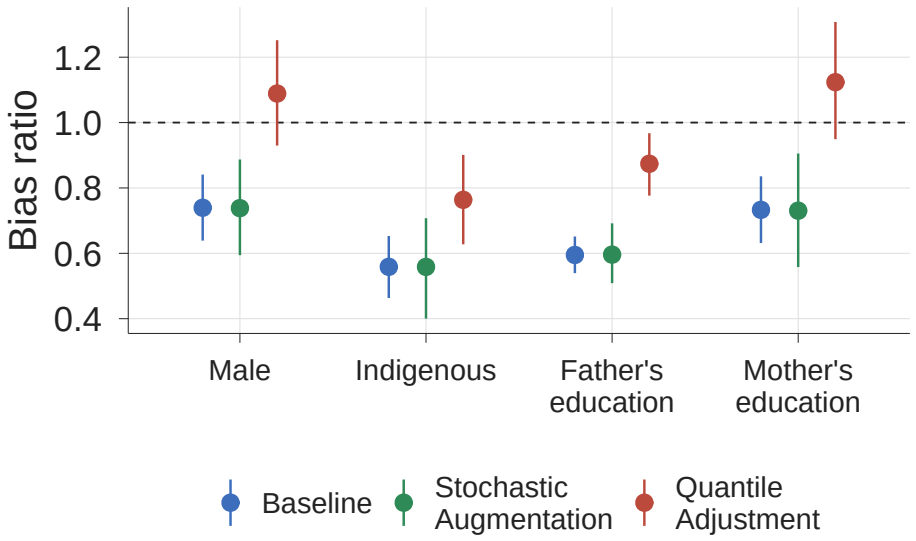



Figure 6: Chile: Bias Ratios for Circumstance Coefficients

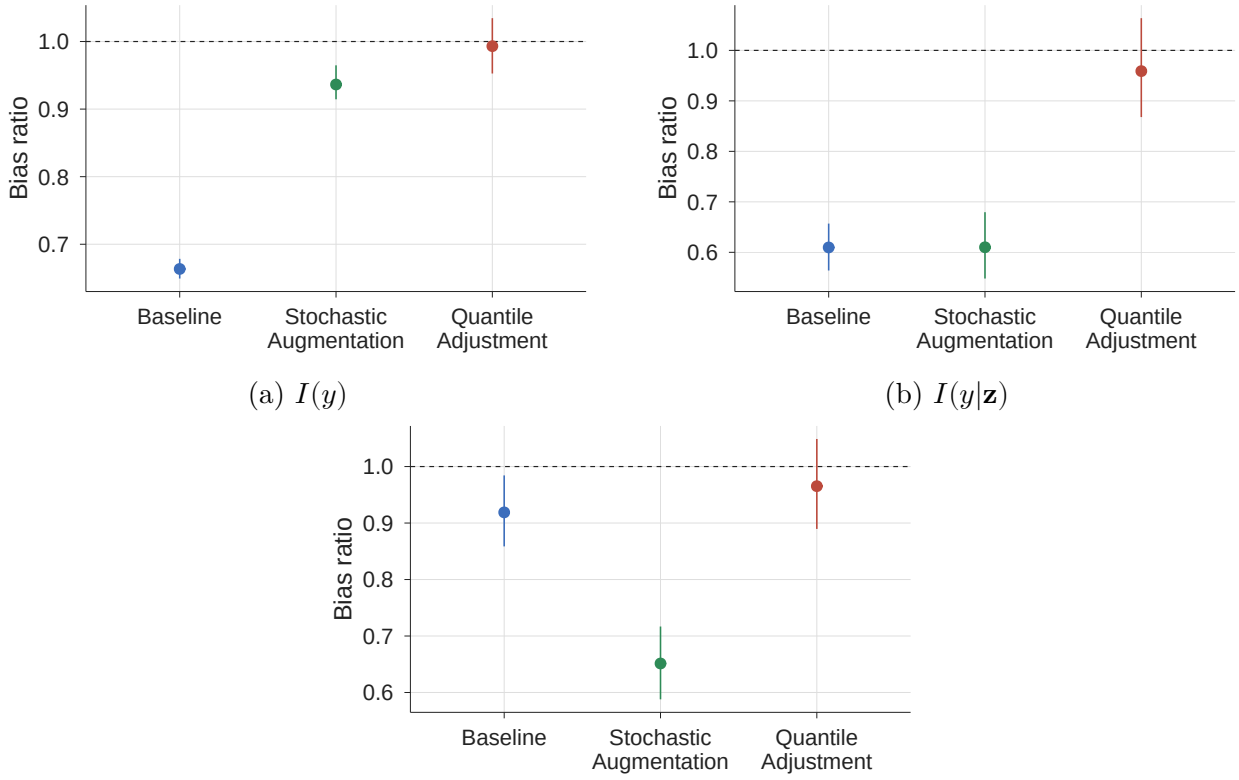
*Notes:* Bias ratio for each circumstance coefficient under the three imputation methods. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. Coefficients are in logs.

$\lambda = 1.002$, and its coefficient still ends 12.4 per cent high, so the error has to come through the off-diagonal weights of Equation (43). Father's education is the likely source. It recovers at $\lambda = 0.828$, and because the two parental indicators are correlated, under-recovering one inflates the coefficient on the other.

Father's education shows the same weighting from the other side. Its covariance is 6.6 per cent short and its coefficient 12.6 per cent short, consistent with a coefficient that nets out the covariance running through the other circumstances, so that the same shortfall is a larger share of it. Correcting one coefficient at a time would not fix either error.

**Inequality measures.** Baseline imputation understates total inequality by 33.7 per cent. Stochastic augmentation closes most of that gap, to 6.4 per cent. It does not close all of it because the variance it restores is the log-scale variance, and the Gini is a level-scale index that depends on the rest of the distribution. The quantile adjustment comes within 0.7 per cent.

For absolute IOp the two existing methods coincide again, both 39.0 per cent below the benchmark, since variance bias does not reach an estimand built only from the coefficients. The quantile adjustment closes the gap to 4.1 per cent.




(a) $I(y)$ (b) $I(y|\mathbf{z})$

(c) $\theta$

Figure 7: Chile: Bias Ratios for Inequality Measures

*Notes:* Bias ratio for total inequality, absolute IOp and the relative share under the three imputation methods. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th empirical percentiles across 1,000 replicates of the split design described in Section 5. Inequality is the Gini coefficient on the level scale.

**The relative share.** The relative share is the IOp estimand applied studies usually report, and it collects both biases. The quantile adjustment lands 3.5 per cent short of the benchmark, against 8.1 per cent for baseline imputation and 34.9 per cent for stochastic augmentation.

Baseline imputation does better on the share than on either component because its two errors partly cancel. A numerator at 61.0 per cent of the benchmark divided by a denominator at

66.3 per cent gives 92 per cent. Stochastic augmentation restores the denominator without touching the numerator, which removes the cancellation and leaves the share further from the benchmark than doing nothing.

Across the twelve estimands in Table 5, the quantile adjustment is nearer the benchmark than both alternatives on eleven. The exception is the variance, where it trails stochastic augmentation by a fifth of a percentage point.

# 6 Discussion

The applications bear on three questions a researcher faces before imputing income. The first is which method suits the estimand, the second whether agreement between two methods says anything about accuracy, and the third how much of the specification choice the screening step can settle. I take them in turn in Sections 6.1 to 6.3.

## 6.1 Estimands and Method Choice

The two biases reach estimands through different channels, so the method an estimand calls for depends on which channels it has. Distributional functionals such as inequality indices depend on the imputed distribution alone and carry no covariance channel. Regression coefficients with imputed income as the dependent variable depend on covariances and have no separate income-variance channel, since $\mathrm{Var}(\mathbf{Z})$ rather than $\mathrm{Var}(y)$ sits in their denominator. Ratios can carry both, and a method that corrects one channel moves such an estimand in a direction the other channel may partly undo.

In Table 6 I report the relative bias of every estimand, which is the mean deviation from the benchmark as a share of the benchmark, and its relative RMSE, which adds the spread across replicates.

Across the nineteen estimands, the quantile adjustment has the smallest absolute bias on fourteen and the lowest relative RMSE on the same fourteen. The five exceptions are the

Table 6: Relative Bias and Relative RMSE by Method

| | | Relative bias (%) | | | Relative RMSE (%) | | |
|---|---|---|---|---|---|---|---|
| Application | Estimand | Baseline | Stochastic | Quantile | Baseline | Stochastic | Quantile |
| Mexico | $\mathrm{Var}(y)$ | −52.0 | **−0.0** | −0.6 | 52.0 | **1.8** | 1.9 |
| | $\mathrm{Cov}(y, z)$ | −41.8 | −41.8 | **−16.3** | 41.8 | 41.8 | **16.3** |
| | $\rho_{y,z}$ | **−16.0** | −41.8 | −16.1 | **16.0** | 41.8 | 16.1 |
| Ecuador | $\mathrm{Var}(y)$ | −35.2 | **−0.0** | 0.0 | 35.2 | **1.1** | 1.2 |
| | $\mathrm{Cov}(y, z)$ | −21.8 | −21.9 | **−2.9** | 21.9 | 21.9 | **3.3** |
| | $\rho_{y,z}$ | **−2.9** | −21.9 | −2.9 | **3.1** | 21.9 | 3.1 |
| | $\beta_{\mathrm{IGE}}$ | 20.6 | −21.9 | **−2.9** | 20.7 | 21.9 | **3.2** |
| Chile | $\mathrm{Var}(y)$ | −51.9 | **−0.0** | −0.2 | 51.9 | **1.9** | 3.2 |
| | $\mathrm{Cov}(y, \mathrm{Male})$ | −26.9 | −27.0 | **7.7** | 27.4 | 28.0 | **11.5** |
| | $\mathrm{Cov}(y, \mathrm{Indigenous})$ | −42.3 | −42.3 | **−19.5** | 42.5 | 42.8 | **20.3** |
| | $\mathrm{Cov}(y, \mathrm{Father})$ | −37.1 | −37.1 | **−6.6** | 37.2 | 37.3 | **8.4** |
| | $\mathrm{Cov}(y, \mathrm{Mother})$ | −33.6 | −33.7 | **−0.3** | 33.8 | 34.1 | **6.9** |
| | $\delta_{\mathrm{Male}}$ | −26.1 | −26.2 | **8.9** | 26.5 | 27.3 | **12.0** |
| | $\delta_{\mathrm{Indigenous}}$ | −44.1 | −44.1 | **−23.6** | 44.4 | 44.8 | **24.6** |
| | $\delta_{\mathrm{Father}}$ | −40.5 | −40.4 | **−12.6** | 40.6 | 40.7 | **13.5** |
| | $\delta_{\mathrm{Mother}}$ | −26.7 | −27.0 | **12.4** | 27.2 | 28.3 | **15.5** |
| | $I(y)$ | −33.7 | −6.4 | **−0.7** | 33.7 | 6.5 | **2.2** |
| | $I(y\|\mathbf{z})$ | −39.0 | −39.0 | **−4.1** | 39.1 | 39.1 | **6.5** |
| | $\theta$ | −8.1 | −34.9 | **−3.5** | 8.7 | 35.0 | **5.4** |

*Notes:* Relative bias is the mean deviation from the benchmark as a percentage of the benchmark; relative RMSE is the root mean squared deviation on the same scale. Both come from 1,000 replicates of the split design described in Section 5. Bold marks the smallest absolute value in each row for each measure, compared before rounding.

three income variances and the Mexican and Ecuadorian correlations. On the variances, stochastic augmentation restores the moment it targets and does so with less spread across replicates. A likely reason is that it draws from the empirical residual distribution, whereas the quantile adjustment carries the estimation error of the quantile gap and its spline.

On the two correlations, baseline imputation is nearer by at most a tenth of a percentage point. Both methods understate the Mexican correlation by about 16 per cent and the Ecuadorian one by about 3 per cent, so the quantile adjustment offers no improvement on either. I take up why the two coincide in Section 6.2.

On the three variances, the quantile adjustment trails stochastic augmentation by at most 0.6 percentage points in absolute relative bias and 1.3 points in relative RMSE. Against

stochastic augmentation on the Chilean covariances and coefficients, and against both alternatives on absolute IOp, the quantile adjustment is closer to the benchmark by between about 15 and 35 percentage points in absolute relative bias. On the Mexican covariance the difference is 25.5 points, and on the Ecuadorian elasticity it is 17.7 points against baseline imputation and 19.0 against stochastic augmentation.

Against baseline imputation on the relative share the difference is 4.6 points in absolute bias and 3.3 in relative RMSE, since that estimand is where the two errors of baseline imputation partly offset.

The composition of the remaining error also differs across methods (Appendix Table G.15). For baseline imputation, squared bias accounts for at least 86 per cent of the mean squared deviation on every estimand, and for stochastic augmentation at least 90 per cent on every estimand except the three variances. Neither method addresses the covariance channel, and an omission that recurs in every replicate does not average out.

Under the quantile adjustment the share depends on how close the relevant recovery fraction is to one. It falls below a tenth on five estimands (the three variances, the covariance with mother's education and total inequality) and stays above 90 per cent on the Mexican covariance and correlation and on both indigenous estimands, where the adjustment recovers only part of the omitted covariance. A larger sample would shrink the variation across replicates for every method but would leave the bias in place, and bias is what dominates the error of baseline imputation and stochastic augmentation.

These results point to the estimand as the basis for choosing a method. Where only the variance of log income matters, stochastic augmentation suffices. Where the estimand depends on a covariance, on the shape of the distribution, or on both, the quantile adjustment performed better in these applications, and where it did not, the loss was small. The one larger loss in the paper comes from the level-scale correlation of Appendix D.1, which I discuss in Section 6.2.

## 6.2 Rescaling and Ratio Estimands

Baseline imputation and the quantile adjustment both understate the Mexican correlation by about 16 per cent and the Ecuadorian one by about 3 per cent, within a tenth of a percentage point of each other in both cases. A researcher who saw two methods agree this closely might take it as a sign that both are close to the truth. In Appendix C I show when such agreement arises and what it implies.

Suppose the observed and predicted distributions have the same shape and differ only in spread. They share a mean, since $\mathbb{E}[\varepsilon \mid \hat{y}] = 0$, so the quantile adjustment then reduces to a rescaling that stretches the predicted distribution about its mean by the factor $1/r$, where $r = \sigma_{\hat{y}}/\sigma_y$ (Equation (C.13)). The stretch multiplies the covariance with $z$ and the standard deviation of imputed income by the same factor, so the correlation does not change (Equation (C.15)). The recovery fraction is then pinned down by the first-stage fit and by how much of the covariance the predictors already capture (Equation (C.14)).

The Mexican log-income distributions are close to that case. On the full working sample, the shape term of Appendix C has a standard deviation of 4 per cent of that of income. With $r = 0.692$ and $\mathrm{Cov}(\hat{y}, z)/\mathrm{Cov}(\varepsilon, z) = 1.394$, the same-shape formula gives $\lambda = 0.620$, against a realised 0.612. The adjustment therefore moves the covariance substantially, cutting its attenuation from 41.8 to 16.3 per cent, and leaves the correlation almost where baseline imputation put it.

When the shapes differ, the correlation can move. In Appendix C I split the adjustment into the rescaling term and a shape term $R(\hat{y})$, and show that the change in the correlation equals $\mathrm{Cov}(R(\hat{y}), z)/(\sigma_y \sigma_z)$ (Equation (C.16)). The size of that change is bounded by how far apart the two shapes are and by how much of $z$ varies with predicted income (Equation (C.17)), so a difference in shape is necessary for the correlation to move but not sufficient.

Ecuador shows the difference between the two. Its quantile gap is not monotone (Appendix Figure E.6), and the shape term has a standard deviation of 24 per cent of that of parental income, more than five times the Mexican figure. The term is positive at both ends of the

distribution and negative in between, so its covariance with offspring income, which rises steadily with predicted parental income, nets out, and the correlation between the two is $-0.0002$.

The correlation therefore stays where baseline imputation put it, and with $r = 0.805$ and a covariance ratio of 3.580 the same-shape formula still gives 0.867, against a realised $\lambda$ of 0.868. The adjustment cuts the covariance attenuation from 21.8 to 2.9 per cent.

The level-scale version of the Mexican exercise shows the other case. There I estimate the gap in pesos, and the adjustment recovers 67.2 per cent of the covariance against 42.0 per cent for baseline imputation, but it ends 32.9 per cent below the benchmark correlation, against 28.5 per cent for baseline imputation (Appendix Table D.4). By Equation (C.16), a correlation that moves implies that the two level-scale distributions differ in shape, and here the difference moves the correlation away from the benchmark.

The shape comparison has a practical use. Both distributions are available in the donor sample, so a researcher can standardise them and compare their shapes before imputing. If the shapes coincide, the adjustment will act on a correlation as a rescaling and leave it unchanged. The comparison does not involve the external variable, so it does not depend on the target sample, but for the same reason it cannot say whether a correlation will move when the shapes differ, as Ecuador shows, or in which direction.

The 16 per cent of baseline imputation has a separate reading, as a 41.8 per cent covariance attenuation working against a compressed denominator. That reading is correct, but the near-cancellation it describes depends on the relative size of two errors that a researcher cannot measure without a benchmark, whereas she can make the shape comparison beforehand.

Agreement between methods is therefore not evidence of accuracy. The Chilean relative share makes the same point from the other side, since baseline imputation comes within 8.1 per cent of the benchmark while the two components it divides are off by 33.7 and 39.0 per cent.

## 6.3 Scope and Limits of the Screening Rule

I show in Section 3.2 that the adjusted covariance is closer to the truth than baseline imputation whenever $0 < \lambda < 2$ (Equation (24)). Overshooting becomes a failure only past $\lambda = 2$, and the screening step of Section 4.4 aims to keep the specification away from that region.

In the Chilean application, retaining the two gender-related predictors puts the respondent's sex at $\lambda = 2.62$, past that point. The screen removes them on the covariance-to-income contribution ratio of Equation (32), which needs no benchmark, and the recovery fraction falls to 1.28. Across replicates the corresponding means are 2.925 and 1.340 (Appendix Table F.13).

Without the screen, the adjusted covariance with sex ends about 25 per cent above the benchmark, further from it than baseline imputation at about 17 per cent below. With the screen, the adjustment still overshoots, but the covariance ends 7.7 per cent from the benchmark, where baseline imputation leaves it 26.9 per cent away. The coefficient on sex moves from 28 to 9 per cent above the benchmark (Appendix Table F.14).

The effect of the screen is concentrated on the respondent's sex. For the other three circumstances the recovery fractions change little, and on the IOp aggregates the unscreened specification is slightly closer to the benchmark. Its relative share is 0.332 against 0.323 with screening, and absolute IOp 0.158 against 0.154, for benchmarks of 0.334 and 0.161. The screen therefore trades a small loss on the aggregates for a large gain on the circumstance it flags.

The screen detects a predictor that explains the external variable while adding little to the prediction of income, which is a property of the specification that the researcher can compute. A small residual dependence between income and the external variable is a different matter, a feature of the two variables themselves, and the respondent's sex after screening illustrates it. Household income is pooled, so sex carries little information about it once household composition is in the specification, and the omitted covariance is small enough that the

adjustment overfills it.

No surviving predictor has a large ratio against sex, the largest being 0.12 for the share of occupied members (Appendix Table F.11), so the screen gives no reason to drop another one.

The screen is therefore one input to the specification rather than a decision rule. It narrows the specification on evidence the data provide, and the remaining judgement, whether the residual dependence is large enough for the adjustment to help, rests on theory and prior evidence, as in the four Chilean circumstances of Section 5.3.2.

# 7 Conclusion

When income and another variable sit in separate samples, researchers predict income from the characteristics both samples share and use the prediction in its place. The prediction is less dispersed than income. It also carries only the part of the relationship between income and the external variable that runs through those characteristics.

Existing corrections deal with these two distortions one at a time and act on the estimand, so each has to be derived again for every new estimand. The quantile adjustment acts on the imputed values instead. It replaces each prediction with the observed income at the same rank, which addresses both distortions in one step and leaves the second stage unchanged.

The adjustment restores the distribution of income by construction, up to estimation error. Its effect on the covariance with the external variable is less automatic. The adjustment depends only on the common predictors, so it can supply only covariance that runs through them.

How much of the missing covariance it puts back depends on how the predictors and the unpredictable part of income relate to the external variable. When the two relate to it in the same direction, the adjustment moves the estimate toward the truth. It improves on baseline imputation even when it overshoots, provided it puts back less than twice the

missing covariance. When they relate to it in opposite directions, the adjustment makes the estimate worse.

Because the outcome depends on a relationship the researcher cannot observe, the method comes with two diagnostics. A screen flags predictors that push the adjustment toward overshooting, and a threshold correlation can be weighed against theory and prior evidence.

In the applications, the adjustment comes closest to the full-sample benchmark on 14 of the 19 estimands, and stays within 0.6 percentage points of the best method on the other 5. Its gains are largest where a covariance or the shape of the distribution matters, as in the Mexican income–consumption covariance, the Ecuadorian intergenerational elasticity and the Chilean inequality of opportunity estimates.

The regime judgements I made from the threshold before observing the recovery fractions held wherever I made one. The screen kept the one case that overshoots, on average, within the range where the adjustment still improves on baseline imputation.

For practice, the choice of method should follow the estimand. Where only the variance of log income matters, stochastic augmentation suffices. Where the estimand involves a covariance or the shape of the distribution, the quantile adjustment performed better in these applications, and the screen and the threshold are part of using it rather than optional checks.

Agreement between two methods is no sign that either is accurate. Baseline imputation can land close to the truth when its two distortions happen to offset.

The main question the paper leaves open is inference. The intervals I report describe how much estimates vary across splits of a single survey, which a researcher with one donor and one target sample cannot reproduce. Standard errors for estimates built on the adjusted values would have to reflect estimation of the first stage and of the quantile gap, as well as sampling in both samples.

The diagnostic needs the same treatment. The regime call currently compares a point esti-

mate of the threshold correlation with a judgement about the residual correlation. Attaching uncertainty to that comparison would tell a researcher how confident she can be that the adjustment helps.

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# Guide to the Appendices

The appendices fall into two groups. In the first three I develop results that the main text relies on. In Appendix A I prove the two sign results of Section 3.2, in Appendix B I show when the sign of the predictor channel can be read from regression coefficients, and in Appendix C I explain when the quantile adjustment leaves a correlation unchanged, which I use in Section 6.2 to interpret the Mexican and Ecuadorian results.

In Appendices D, E and F I collect supplementary material for the three applications of Section 5, in the same order. Each follows the same sequence. It starts with the working sample, then reports the screening of predictors and the regime diagnostics, and shows the estimated quantile gap and the income distributions it produces. Each ends with an alternative specification, the estimands in levels for Mexico, the rank-rank slope for Ecuador and the unscreened predictor set for Chile. In Appendix G I split the error of each method into bias and variation across replicates.

I collect in the table below the notation the appendices use, all of which the main text defines.

| Symbol | Meaning |
|---|---|
| $y$ | Income, in logs in the applications |
| $\mathbf{X}$ | Predictors observed in both samples |
| $\hat{y} = \mathbb{E}[y \mid \mathbf{X}]$ | Predicted income |
| $\varepsilon = y - \hat{y}$ | Prediction error |
| $R^2$, $r = \sigma_{\hat{y}}/\sigma_y = \sqrt{R^2}$ | First-stage fit, ratio of standard deviations |
| $z$ | External variable, observed in the target sample |
| $g(\mathbf{X}) = \mathbb{E}[z \mid \mathbf{X}]$, $z_\perp = z - g(\mathbf{X})$ | Explained and unexplained parts of $z$ |
| $\bar{g}(t) = \mathbb{E}[g(\mathbf{X}) \mid \hat{y} = t]$ | Expected $z$ given predicted income $t$ |
| $Q_y(p)$, $Q_{\hat{y}}(p)$, $F_{\hat{y}}$ | Quantile functions, distribution function of $\hat{y}$ |
| $\eta(p) = Q_y(p) - Q_{\hat{y}}(p)$ | Quantile gap at rank $p$ |
| $\tilde{\eta} := \eta(p)$, $p = F_{\hat{y}}(\hat{y})$, $\tilde{y}_q = \hat{y} + \tilde{\eta}$ | Adjustment at own rank, adjusted income |
| $\lambda = \mathrm{Cov}(\tilde{\eta}, z)/\,\mathrm{Cov}(\varepsilon, z)$ | Recovery fraction |
| $\rho = \mathrm{Corr}(\varepsilon, z_\perp)$, $\rho^*$ | Residual correlation, exact-recovery threshold |
| $S_i(z_k)$ | Contribution ratio of predictor $i$ for $z_k$ |

# A Sign of the Quantile Adjustment

Section 3.2 uses two sign results. Equation (13) states that the adjustment is larger for observations with higher predicted income, $\text{Cov}(\hat{y}, \tilde{\eta}) \geq 0$. Equation (21) states that the predictor channel is positive when the expected external variable rises with predicted income, $\text{Cov}(\tilde{\eta}, g(\mathbf{X})) \geq 0$ whenever $\bar{g}$ is nondecreasing. Both follow from the rearrangement inequality.

As in Section 2.1, $\hat{y} = \mathbb{E}[y \mid \mathbf{X}]$, and $U := F_{\hat{y}}(\hat{y})$ is an observation's rank in predicted income. Because $\hat{y}$ is continuously distributed, $U$ is uniform on $[0, 1]$ and $Q_y(U)$ has the same distribution as observed income. The adjustment is

$$\tilde{\eta} = Q_y(U) - \hat{y}. \tag{A.1}$$

Let $h$ be a nondecreasing function for which $h(\hat{y})$ has finite variance. Both $h(\hat{y})$ and $Q_y(U)$ are nondecreasing in $U$, and $Q_y(U)$ has the same distribution as $y$, so the two variables are paired in the order that maximises their covariance (Tchen 1980). This gives $\text{Cov}(h(\hat{y}), Q_y(U)) \geq \text{Cov}(h(\hat{y}), y)$, and therefore

$$\begin{aligned} \text{Cov}\left(h(\hat{y}), \tilde{\eta}\right) &= \text{Cov}\left(h(\hat{y}), Q_y(U)\right) - \text{Cov}\left(h(\hat{y}), \hat{y}\right) \\ &\geq \text{Cov}\left(h(\hat{y}), y\right) - \text{Cov}\left(h(\hat{y}), \hat{y}\right) \\ &= 0. \end{aligned} \tag{A.2}$$

The last equality holds because $\hat{y}$ is the conditional mean of $y$ given the predictors, so $y - \hat{y}$ is uncorrelated with any function of $\hat{y}$.

Taking $h(t) = t$ gives $\text{Cov}(\hat{y}, \tilde{\eta}) \geq 0$, which is Equation (13).

By Equation (20), $\text{Cov}(\tilde{\eta}, g(\mathbf{X})) = \text{Cov}(\tilde{\eta}, \bar{g}(\hat{y}))$. The variance of $\bar{g}(\hat{y})$ cannot exceed that of $z$, so taking $h = \bar{g}$ gives $\text{Cov}(\tilde{\eta}, g(\mathbf{X})) \geq 0$ when $\bar{g}$ is nondecreasing, which is Equation (21). Applying the same argument to $-\bar{g}$ reverses the sign when $\bar{g}$ is nonincreasing.

For strictly increasing $h$, the inequality in Equation (A.2) is strict unless $y$ is already ordered by $\hat{y}$. Were it so ordered, $y$ would be a nondecreasing function of $\hat{y}$ and therefore measurable with respect to the predictors, giving $y = \mathbb{E}[y \mid \mathbf{X}] = \hat{y}$ and $R^2 = 1$. Any first stage with $R^2 < 1$ therefore yields $\text{Cov}(\hat{y}, \tilde{\eta}) > 0$, and a strictly increasing $\bar{g}$ yields a strictly positive predictor channel.

The last equality in Equation (A.2) is the only place where the prediction enters, so the argument requires $\hat{y}$ to be the conditional mean of $y$. It also requires $\hat{y}$ to be continuous, since $Q_y(U)$ otherwise fails to reproduce the distribution of $y$, and mass points in $\hat{y}$ break it for the reason I give in Section 4.3. Nothing restricts the shape of either distribution or requires $\eta(p)$ to be increasing, as the Ecuadorian application illustrates (Figure E.6). The result concerns population quantities and holds up to estimation error with an estimated first stage and a smoothed $\hat{\eta}(p)$.

# B Sign of the Predictor Channel Under Linearity

Equation (21) ties the sign of the predictor channel to whether $\bar{g}$ rises or falls with predicted income, which is hard to check in an application because $\bar{g}$ is a conditional expectation rather than something a regression reports. Under the conditions below the sign follows instead from quantities a researcher already estimates, namely the first-stage coefficients $\beta$, the coefficients $\gamma$ from regressing $z$ on the predictors, and the covariance matrix of the predictors.

## The Sign From the Coefficients

I use three conditions,

$$\text{(i)} \quad \hat{y} = \beta' \mathbf{X}, \tag{B.3}$$

$$\text{(ii)} \quad g(\mathbf{X}) = \gamma' \mathbf{X}, \tag{B.4}$$

$$\text{(iii)} \quad \bar{g}(t) \text{ is linear in } t. \tag{B.5}$$

Condition (i) says that predicted income is a linear combination of the predictors, and condition (ii) says the same of the part of $z$ the predictors explain. Condition (iii) says that, among observations sharing the same predicted income $t$, the average of that explained part changes along a straight line as $t$ changes.

Under (ii),

$$\bar{g}(t) = \gamma' \, \mathbb{E}[\mathbf{X} \mid \beta' \mathbf{X} = t], \tag{B.6}$$

so condition (iii) concerns the distribution of $\mathbf{X}$ rather than the coefficients, and no restriction on $\beta$ or $\gamma$ delivers it. Joint normality of $\mathbf{X}$ is sufficient, since any two linear combinations of a jointly normal vector are bivariate normal and a bivariate normal has a linear conditional expectation (Anderson 2003).[11] With a single predictor and $\beta \neq 0$, conditioning on $\beta x = t$

[11] The property extends to the elliptical family, the class of distributions whose density contours are ellipses, which includes the normal and the multivariate $t$ (Cambanis et al. 1981).

fixes $x$ at $t/\beta$, so Equation (B.6) gives $\bar{g}(t) = \gamma t/\beta$ and (iii) holds with no further assumption. The predictors in the applications of Section 5 are mostly categorical, so there (iii) holds only approximately.

Under (i) to (iii), $\bar{g}$ is the linear projection of $g(\mathbf{X})$ on $\hat{y}$, with slope

$$\psi = \frac{\gamma'\Sigma\beta}{\beta'\Sigma\beta}, \qquad \Sigma := \mathrm{Var}(\mathbf{X}), \tag{B.7}$$

the covariance between expected $z$ and predicted income divided by $\mathrm{Var}(\hat{y})$, which is strictly positive under the assumption $R^2 > 0$ of Section 2.2. Substituting into Equation (20) and using the linearity of covariance,

$$\mathrm{Cov}\left(\tilde{\eta}, g(\mathbf{X})\right) = \mathrm{Cov}\left(\tilde{\eta}, \bar{g}(\hat{y})\right) = \psi\,\mathrm{Cov}(\hat{y}, \tilde{\eta}). \tag{B.8}$$

I show in Appendix A that the co-movement is strictly positive whenever $R^2 < 1$, so for $0 < R^2 < 1$ the channel takes the sign of $\psi$,

$$\mathrm{sign}\left[\mathrm{Cov}\left(\tilde{\eta}, g(\mathbf{X})\right)\right] = \mathrm{sign}\left[\gamma'\Sigma\beta\right]. \tag{B.9}$$

With a single predictor $x$, $\Sigma$ reduces to $\sigma_x^2$ and the numerator becomes $\gamma\beta\sigma_x^2$, so the channel takes the sign of the product $\gamma\beta$ rather than of $\gamma$ alone. A predictor that lowers income ($\beta < 0$) and lowers $z$ ($\gamma < 0$) contributes positively, since it moves observations down the predicted distribution and down $g(\mathbf{X})$ together. With several predictors, correlation among them contributes through the off-diagonal elements of $\Sigma$, so the bilinear form $\gamma'\Sigma\beta$ settles the sign and individual coefficients do not.

## The Sign and the Recovery Regimes

From Equations (18) and (B.8),

$$\lambda = \frac{\psi\,\mathrm{Cov}(\hat{y}, \tilde{\eta})}{\mathrm{Cov}(\varepsilon, z)}, \tag{B.10}$$

so mis-recovery ($\lambda < 0$) occurs exactly when $\gamma'\Sigma\beta$ and $\mathrm{Cov}(\varepsilon, z)$ have opposite signs. In the single-predictor case, $\gamma\beta$ has the opposite sign to the residual dependence between income and $z$.

That configuration requires the predictors and the channels they miss to track $z$ in opposite directions. Suppose the target survey records a wife's earnings $z$ with her husband's characteristics, and the researcher imputes his earnings $y$ from his education, age and region. People tend to marry partners with similar education (Eika et al. 2019), so men whose characteristics predict high earnings tend to have wives who earn more, and the predictor channel is positive. Within a group of observationally similar men, however, married women's hours tend to fall when their husbands' wages rise (Devereux 2004; Blau and Kahn 2007). Where that response dominates, $\lambda < 0$ and the adjustment moves an already overstated covariance further up.

The line between partial covariance adjustment and over-recovery depends on the magnitude of $\lambda$ rather than its sign, so the coefficients leave it open and the comparison of Section 4.4 settles it.

## Estimation and Departures from Linearity

Every term in Equation (B.9) can be estimated. The first-stage coefficients $\hat{\beta}$ come from the donor sample and $\hat{\gamma}$ from regressing $z$ on $\mathbf{X}$ in the target sample, which Section 4.4 already requires for the denominator of $\rho^*$. Under the common-population assumption of Section 2 either sample estimates $\Sigma$, and where I relax it in Section 4.2 the target sample or a reweighted donor sample is the right one, since the sign result concerns the population to which she applies the adjustment.

Without (iii), $\psi$ is still the slope of the best straight-line approximation to $\bar{g}$. The part of $\bar{g}$ that the straight line misses is uncorrelated with $\hat{y}$ but not with $\tilde{\eta}$, which is a nonlinear function of $\hat{y}$, and nothing bounds its contribution, so Equation (B.9) is no longer guaranteed. The sign result of Equation (21) does not need (iii) and still holds whenever $\bar{g}$ is increasing

(Appendix A).

Once $\tilde{\eta}$ is constructed and $z$ regressed on the predictors, $\mathrm{Cov}(\tilde{\eta}, g(\mathbf{X}))$ can be estimated directly, with no linearity anywhere. The coefficient result therefore earns its place in the regime analysis rather than in estimating the channel. Equation (B.9) gives the sign of the numerator of Equation (B.10), and mis-recovery turns on whether that sign matches the sign of $\mathrm{Cov}(\varepsilon, z)$, which Section 4.4 rather than the coefficients has to settle.

# C Shape Differences and the Correlation

In Section 6.2 I report that baseline imputation and the quantile adjustment return almost the same correlation in the Mexican application. When the observed and predicted income distributions have the same shape and differ only in spread, the adjustment stretches predicted income by a constant factor, which fixes the recovery fraction and leaves any correlation unchanged. When the shapes differ, the correlation can move, by an amount this appendix bounds.

## The Stretch Under Equal Shapes

Since $\hat{y} = \mathbb{E}[y \mid \mathbf{X}]$, the observed and predicted distributions share the mean $\mu := \mathbb{E}[y]$. To compare their shapes, remove the mean and divide by the standard deviation, which gives the standardised quantile functions

$$Q_y^s(p) := \frac{Q_y(p) - \mu}{\sigma_y}, \qquad Q_{\hat{y}}^s(p) := \frac{Q_{\hat{y}}(p) - \mu}{\sigma_{\hat{y}}}, \tag{C.11}$$

and let $r := \sigma_{\hat{y}}/\sigma_y$ be the ratio of standard deviations, which the variance decomposition of Section 2.2 makes equal to $\sqrt{R^2}$. The two distributions have the same shape when $Q_y^s = Q_{\hat{y}}^s$, which I write as $Q^s$. A pair of normal distributions satisfies this, as does any pair in which one distribution is a shifted and stretched version of the other.

Under that condition the quantile gap of Equation (9) is proportional to $Q^s$,

$$\eta(p) = [\mu + \sigma_y Q^s(p)] - [\mu + \sigma_{\hat{y}} Q^s(p)] = (\sigma_y - \sigma_{\hat{y}})\, Q^s(p). \tag{C.12}$$

At an observation's own rank $p = F_{\hat{y}}(\hat{y})$, $Q^s(p) = (\hat{y} - \mu)/\sigma_{\hat{y}}$, so

$$\tilde{\eta} = \frac{1 - r}{r}\,(\hat{y} - \mu), \qquad \tilde{y}_q = \mu + \frac{1}{r}\,(\hat{y} - \mu). \tag{C.13}$$

The adjustment stretches the predicted distribution about its mean by the factor $1/r$. This

restores the variance, since $\mathrm{Var}(\tilde{y}_q) = \sigma^2_{\hat{y}}/r^2 = \sigma^2_y$.

The stretch also fixes the recovery fraction. Taking covariances with $z$ in Equation (C.13) and substituting into Equation (14),

$$\lambda = \frac{1-r}{r} \cdot \frac{\mathrm{Cov}(\hat{y}, z)}{\mathrm{Cov}(\varepsilon, z)}, \tag{C.14}$$

so the share of the omitted covariance the adjustment restores depends only on the first-stage fit, through $r$, and on how much of the covariance the predictors already capture. In the Mexican application, $r = 0.692$ and $\mathrm{Cov}(\hat{y}, z)/\mathrm{Cov}(\varepsilon, z) = 1.394$, so Equation (C.14) gives $\lambda = 0.620$, against a realised 0.612. In the Ecuadorian application, $r = 0.805$ and the ratio is 3.580, so the formula gives 0.867, against a realised 0.868 (Table E.7).

The adjustment multiplies the covariance with $z$ by $1/r$ and the standard deviation of imputed income by the same factor, so their ratio is unchanged and

$$\mathrm{Corr}(\tilde{y}_q, z) = \mathrm{Corr}(\hat{y}, z). \tag{C.15}$$

In the Mexican application, the two correlations come within a tenth of a percentage point of each other, and in the Ecuadorian application within a hundredth.

## Correlation Under Differing Shapes

When the distributions differ in shape, the adjustment does more than stretch predicted income. To isolate that part, compare baseline and adjusted income after standardising each by its own standard deviation. The adjustment reproduces the observed income distribution, so $\sigma_{\tilde{y}_q} = \sigma_y$ and each correlation is the covariance of two standardised variables. Subtracting one from the other,

$$\mathrm{Corr}(\tilde{y}_q, z) - \mathrm{Corr}(\hat{y}, z) = \mathrm{Cov}\left(\frac{\tilde{y}_q - \mu}{\sigma_y} - \frac{\hat{y} - \mu}{\sigma_{\hat{y}}},\ \frac{z}{\sigma_z}\right). \tag{C.16}$$

The identity is exact, and it requires only the Berkson structure of Section 2.1 and continuity of $\hat{y}$.

The difference inside the covariance measures how much the adjustment changes income at each predicted-income rank, beyond a change in scale. When the two standardised quantile functions of Equation (C.11) coincide, that difference is zero at every rank, so any change in the correlation comes from a difference in shape rather than in spread.

The standardised difference also has to covary with $z$. Both imputed incomes are functions of predicted income, so by the argument of Section 3.2 their difference reaches $z$ only through $\bar{g}(\hat{y})$, the expected value of $z$ at each predicted income. The Cauchy–Schwarz inequality then gives

$$\left|\operatorname{Corr}(\tilde{y}_q, z) - \operatorname{Corr}(\hat{y}, z)\right| \leq \underbrace{\operatorname{SD}\left(\frac{\tilde{y}_q - \mu}{\sigma_y} - \frac{\hat{y} - \mu}{\sigma_{\hat{y}}}\right)}_{\text{Difference in shape}} \cdot \underbrace{\frac{\sigma_{\bar{g}(\hat{y})}}{\sigma_z}}_{\text{Spread of expected } z} . \tag{C.17}$$

Each factor is necessary, and the two together are not sufficient, since the bound caps the change without setting it.

The Mexican and Ecuadorian applications stay stable for different reasons. In Mexico the standard deviation of the standardised difference is 0.04, so the distributions are close in shape and the first factor leaves little room for the correlation to move. In Ecuador it is 0.24, and the second factor is at least $\operatorname{Corr}(\hat{y}, z) = 0.403$, so Equation (C.17) would permit a change of around ten percentage points. The quantile gap there is not monotone (Figure E.6), and the standardised difference is positive at both ends of the distribution and negative in between, so its covariance with offspring income cancels. Its correlation with offspring income is $-0.0002$, and the adjusted correlation stays where baseline imputation put it. The bound cannot see that cancellation, which is why Ecuador sits far inside it.

## The Scale of the Adjustment

These results apply to correlations computed on the scale used to estimate the quantile gap. A correlation computed on another scale can behave differently. A constant stretch of log income becomes a nonlinear transformation once converted back to levels, for example, so it can change the correlation in levels even where the log-scale correlation is unchanged. In Appendix D.1 I estimate the gap in levels instead, where the same reasoning applies to the level-scale distributions.

# D Joint Distribution: Supplementary Material

Here I collect supplementary material for the Mexican application of Section 5.1, which correlates household income with consumption. The first part follows the exercise of the main text, with both variables in logs. Appendix D.1 repeats it with income and consumption in pesos, and I use its correlation result in Section 6.2.

## Working Sample

Table D.1: Mexico: Working Sample

| Variable | Mean | SD | Variable | Mean | SD |
|---|---|---|---|---|---|
| Log income | 9.885 | 0.725 | Dwelling type: 5 | 0.004 | 0.063 |
| Log consumption | 9.339 | 0.740 | Head education: 1 | 0.052 | 0.222 |
| Household size | 3.355 | 1.741 | Head education: 2 | 0.000 | 0.015 |
| Share female | 0.519 | 0.265 | Head education: 3 | 0.118 | 0.322 |
| Share occupied | 0.525 | 0.315 | Head education: 4 | 0.150 | 0.357 |
| Share receiving income | 0.728 | 0.268 | Head education: 5 | 0.027 | 0.163 |
| Locality size: 1 | 0.496 | 0.500 | Head education: 6 | 0.259 | 0.438 |
| Locality size: 2 | 0.153 | 0.360 | Head education: 7 | 0.033 | 0.179 |
| Locality size: 3 | 0.146 | 0.353 | Head education: 8 | 0.152 | 0.359 |
| Locality size: 4 | 0.204 | 0.403 | Head education: 9 | 0.040 | 0.196 |
| Dwelling type: 1 | 0.137 | 0.344 | Head education: 10 | 0.139 | 0.346 |
| Dwelling type: 2 | 0.613 | 0.487 | Head education: 11 | 0.030 | 0.170 |
| Dwelling type: 3 | 0.240 | 0.427 | Head gender: 1 | 0.665 | 0.472 |
| Dwelling type: 4 | 0.006 | 0.077 | Head gender: 2 | 0.335 | 0.472 |

*Notes:* Weighted means and standard deviations over 91,242 households, representing 38,771,194 households. The first two rows are the variables the correlation relates, and the remaining rows are the candidate predictors. For a categorical predictor, each row gives the share of households in one category. Head education and head gender are the household head's. State fixed effects (32 categories) are omitted.

## Variable Selection

Table D.2: Mexico: Contribution Ratios

| Predictor | Mean $S$ | 2.5th pct | 97.5th pct |
|---|---|---|---|
| Share female | 7.87 | 1.03 | 37.93 |
| Household head gender | 1.42 | 0.35 | 4.15 |
| Household head education | 1.13 | 1.03 | 1.21 |
| Share occupied | 1.05 | 0.68 | 1.41 |
| Dwelling type | 0.89 | 0.73 | 1.04 |
| Household size | 0.70 | 0.61 | 0.78 |
| Locality size | 0.65 | 0.52 | 0.78 |
| State | 0.60 | 0.54 | 0.70 |
| Share receiving income | 0.17 | 0.12 | 0.22 |

*Notes:* Covariance-to-income contribution ratio $S_i(z_k)$ from Equation (32), computed against log consumption for each candidate predictor. The screen removes a predictor whose ratio exceeds the cutoff $\tau = 10$. The percentiles come from 100 subsamples. No mean ratio exceeds the cutoff, and I retain all nine predictors. The 97.5th percentile for the share of female members lies above it.

## Regime Diagnostics

Table D.3: Mexico: Regime Diagnostics

| Quantity | Subsamples | Full sample | Mean | 2.5th pct | 97.5th pct |
|---|---|---|---|---|---|
| $R^2$ | Joint | 0.479 | 0.481 | 0.471 | 0.491 |
| $\rho$ | Joint | 0.557 | 0.555 | 0.545 | 0.567 |
| $\rho^*$ | Joint | 0.341 | 0.341 | 0.333 | 0.348 |
| $\lambda$ | Joint | 0.612 | 0.614 | 0.595 | 0.630 |
| $\mathrm{Cov}(\tilde{\eta}, z)$ | Joint | 0.101 | 0.101 | 0.098 | 0.103 |
| $\mathrm{Cov}(\varepsilon, z)$ | Joint | 0.165 | 0.164 | 0.159 | 0.170 |
| $R^2$ | Split | | 0.480 | 0.471 | 0.489 |
| $\rho^*$ | Split | | 0.340 | 0.330 | 0.350 |
| $\mathrm{Cov}(\tilde{\eta}, z)$ | Split | | 0.100 | 0.097 | 0.104 |

*Notes:* Joint subsamples draw one sample of 30,000 households and use it as both donor and target, which $\rho$ and $\lambda$ require because they need income and consumption for the same households. Split subsamples draw donor and target independently, the setting a researcher faces, where only $\rho^*$ can be computed. The percentiles come from 100 subsamples. Income and consumption in logs.

## Quantile Gap and Imputed Distributions

The estimated quantile gap in Figure D.1 rises steadily across the distribution, and Figure D.2 shows the imputed densities each method produces from it.

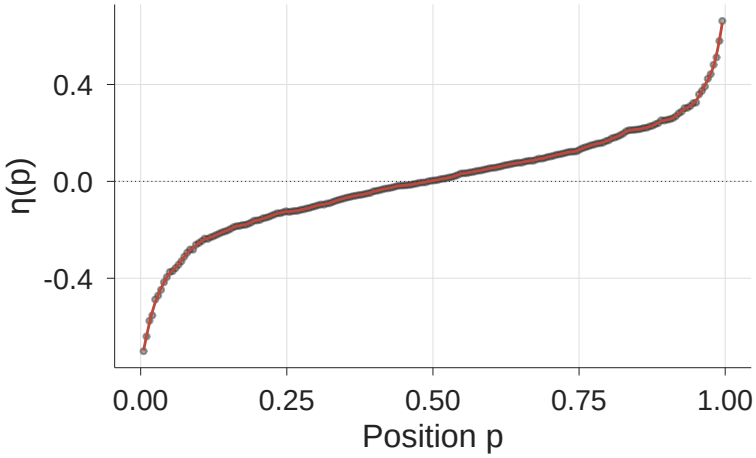



Figure D.1: Mexico: Estimated Quantile Gap

*Notes:* Estimated quantile gap $\hat{\eta}(p)$ from a single replicate. Points give the gap at each of the 200 grid positions and the line is the smoothing spline. At each rank $p$, the gap is the amount the adjustment adds to predicted income, so negative values lower predicted income and positive values raise it. Income in logs.

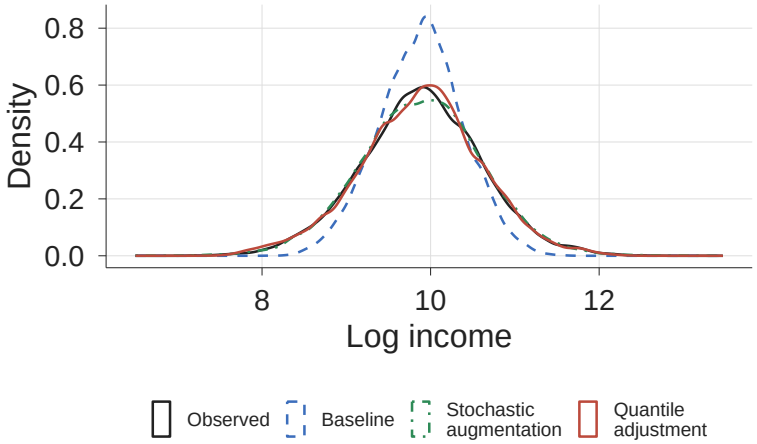



Figure D.2: Mexico: Imputed and Observed Income

*Notes:* Kernel density estimates of imputed log income in the target sample under each method, from the same replicate as Figure D.1, against the observed distribution. A method that restores the distribution of income produces a density close to the observed one.

## D.1 Estimands in Levels

Here I repeat the exercise of Section 5.1 using income and consumption in pesos rather than logs. The covariance decompositions of Section 2.2 require predictions to equal the conditional mean on the scale of analysis, and a first stage estimated in logs does not generally satisfy that requirement in levels.

Suppose the log-scale model is

$$\log y = f(\mathbf{X}) + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid \mathbf{X}] = 0. \tag{D.18}$$

Exponentiating the prediction does not recover the conditional mean in pesos, since

$$\mathbb{E}[y \mid \mathbf{X}] = \exp\big(f(\mathbf{X})\big)\, \mathbb{E}\big[\exp(\varepsilon) \mid \mathbf{X}\big]. \tag{D.19}$$

By Jensen's inequality the second factor exceeds one wherever the conditional variance of the log error is positive, and it may also vary with the predictors. Duan's smearing correction estimates that factor by the average exponentiated residual, so a single correction suffices only where the factor is constant across predictor values.

I therefore estimate the first stage as a generalised linear model with a log link and a Gaussian family,

$$\mathbb{E}[y \mid \mathbf{X}] = \exp\big(\alpha + \mathbf{X}^{\top}\beta\big), \tag{D.20}$$

which targets the conditional mean in pesos directly. Where the mean function is correctly specified the prediction error has conditional mean zero, and the decompositions of Section 2.2 retain their interpretation. I use the predictors the screen retained in the log exercise, and the same first stage serves baseline imputation and stochastic augmentation.

The scale also matters for the quantile adjustment. Estimating the quantile gap in logs and exponentiating the adjusted values reproduces the marginal distribution of income in pesos,

since exponentiation preserves quantile ranks. The adjustment itself becomes multiplicative,

$$\exp(\tilde{y}_q) = \exp(\hat{y}) \exp(\tilde{\eta}), \tag{D.21}$$

so the additive covariance results of Section 3.2 cannot be carried from logs to levels unchanged.

I instead estimate both quantile functions in pesos and apply the adjustment to the level-scale prediction, with $\tilde{\eta} = Q_y(p) - Q_{\hat{y}}(p)$ at an observation's own rank $p = F_{\hat{y}}(\hat{y})$. With $\hat{y}$ the level-scale conditional mean, the decomposition of Section 3.2 applies in pesos, and the recovery fraction, the diagnostic and the regimes keep their interpretation.

Income in pesos is strongly right-skewed, so the quantile gap changes more sharply near the top of the distribution, as Figure D.4 shows. I therefore use $G = 2{,}000$ grid points, against the 200 of the log exercise.

**Results**

The first stage attains $R^2 = 0.350$ [0.330, 0.371], against 0.480 in logs. I report the point estimates in Table D.4 and the bias ratios in Figure D.3.

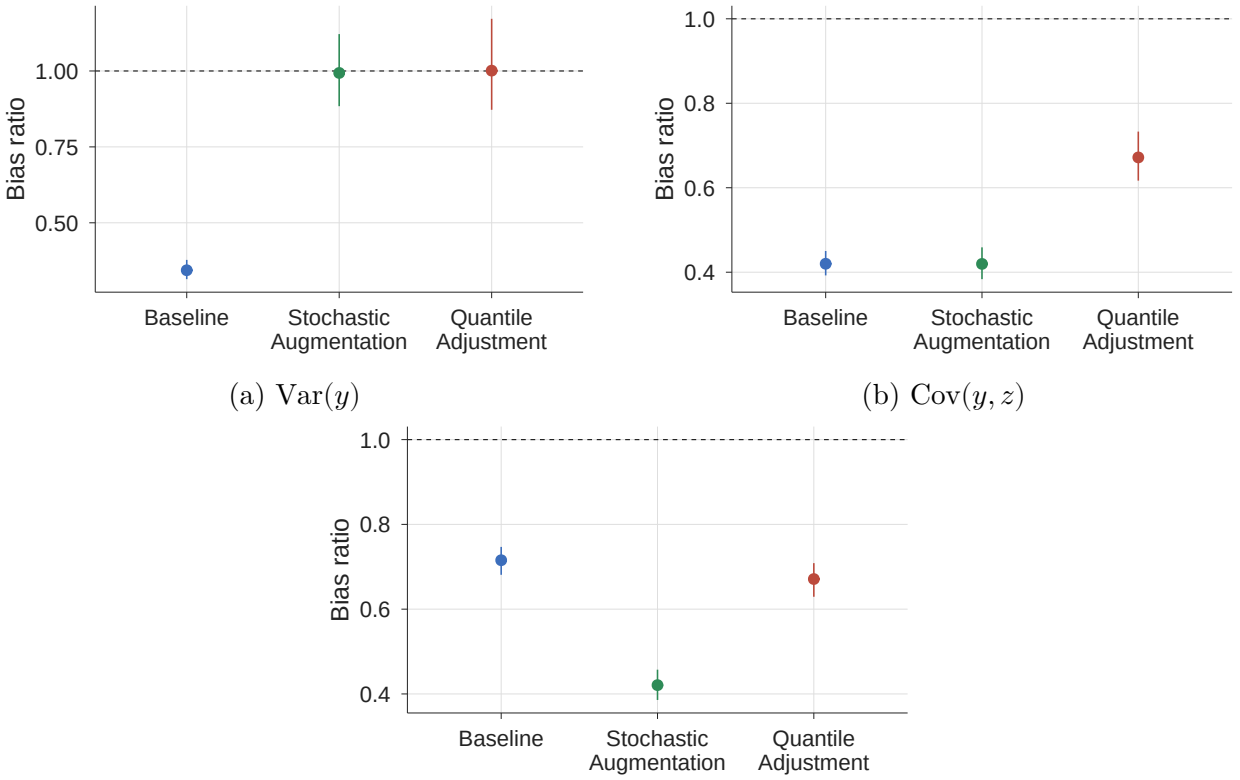
Table D.4: Mexico in Levels: Income–Consumption Estimates

| | Benchmark | Baseline | Stochastic | Quantile |
|---|---|---|---|---|
| $\mathrm{Var}(y)$ | 485.4 | 166.9<br>[152.6, 183.4] | 482.3<br>[429.1, 544.4] | 485.9<br>[423.4, 568.8] |
| $\mathrm{Cov}(y, z)$ | 207.1 | 87.0<br>[81.3, 93.3] | 86.9<br>[79.5, 95.0] | 139.2<br>[127.8, 151.8] |
| $\rho_{y,z}$ | 0.706 | 0.505<br>[0.481, 0.527] | 0.297<br>[0.272, 0.323] | 0.474<br>[0.444, 0.500] |

*Notes:* Benchmark values come from the full working sample. Brackets give the 2.5th and 97.5th percentiles across 1,000 replicates of the split design described in Section 5. Variance and covariance in millions of pesos squared. The first stage is the log-link GLM of Equation (D.20), and the quantile gap is estimated in pesos.

Baseline imputation recovers 34.4 per cent of the variance and 42.0 per cent of the covariance. Stochastic augmentation restores the variance to within 0.6 per cent and leaves the covariance where baseline imputation put it. The quantile adjustment restores the variance to within 0.1 per cent and recovers 67.2 per cent of the covariance.

The correlation ranks the methods differently. Baseline imputation ends 28.5 per cent below the benchmark, the quantile adjustment 32.9 per cent below, and stochastic augmentation 57.9 per cent below. By Equation (C.16), a correlation that moves under the adjustment implies that the two level-scale distributions differ in shape, and here the difference moves the correlation away from the benchmark. I discuss this case in Section 6.2.




(a) $\mathrm{Var}(y)$

(b) $\mathrm{Cov}(y, z)$

(c) $\rho_{y,z}$

Figure D.3: Mexico in Levels: Bias Ratios

*Notes:* Bias ratio for each moment under the three imputation methods. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th percentiles across 1,000 replicates of the split design described in Section 5. Income and consumption in pesos.

**Quantile Gap and Imputed Distributions**

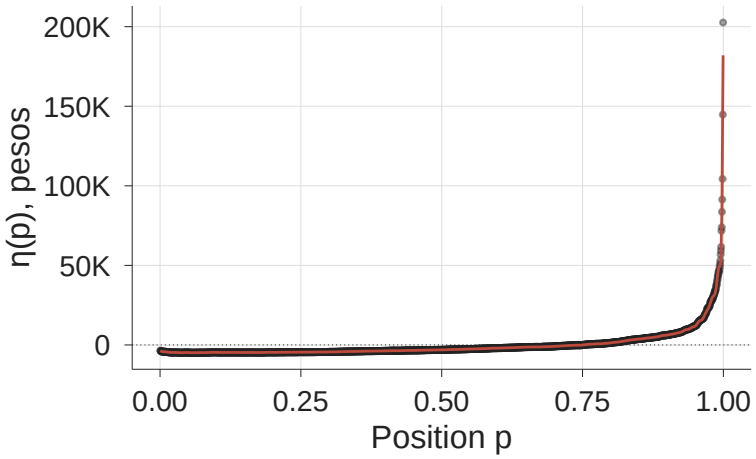



Figure D.4: Mexico in Levels: Estimated Quantile Gap

*Notes:* Estimated quantile gap $\hat{\eta}(p)$ from a single replicate. Points give the gap at each of the 2,000 grid positions and the line is the smoothing spline. At each rank $p$, the gap is the amount the adjustment adds to predicted income. Income in pesos.

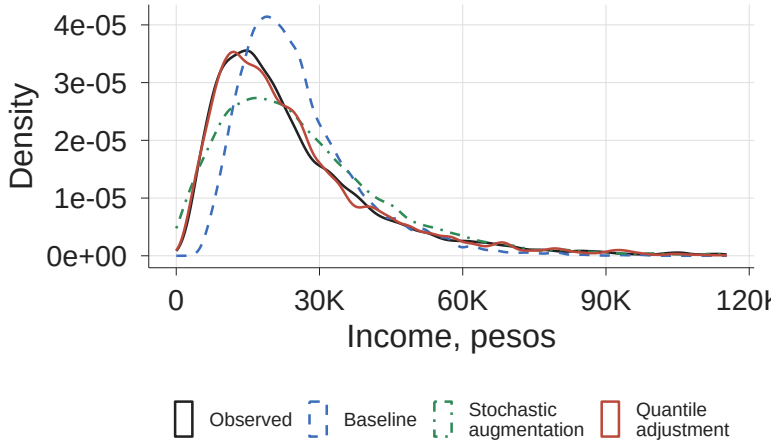



Figure D.5: Mexico in Levels: Imputed and Observed Income

*Notes:* Kernel density estimates of imputed income in the target sample under each method, from the same replicate as Figure D.4, against the observed distribution. A method that restores the distribution of income produces a density close to the observed one.

# E Elasticities: Supplementary Material

Here I collect supplementary material for the intergenerational elasticity of Section 5.2, which relates offspring income to parental income. Appendix E.1 reports the rank-rank slope as an alternative measure of mobility. Both incomes are in logs, and the registers carry no sampling weights, so every quantity here is unweighted.

## Working Sample

Table E.5: Ecuador: Working Sample

| Variable | Mean | SD |
|---|---|---|
| Log parental income | 6.553 | 0.802 |
| Log offspring income | 6.200 | 0.776 |
| Parents' age | 49.521 | 7.803 |
| Male | 0.577 | 0.494 |
| Years of education | 12.062 | 4.558 |
| Indigenous language | 0.089 | 0.285 |
| Sector: Private | 0.544 | 0.498 |
| Sector: Public | 0.150 | 0.357 |
| Sector: Peasant social security | 0.160 | 0.367 |
| Sector: Domestic employment | 0.075 | 0.263 |
| Sector: Voluntary | 0.051 | 0.220 |
| Sector: Semi-contributory | 0.020 | 0.140 |
| Region: Coast | 0.462 | 0.499 |
| Region: Highlands | 0.444 | 0.497 |
| Region: Amazon | 0.045 | 0.207 |
| Region: Galápagos | 0.006 | 0.077 |
| Region: Not delimited | 0.004 | 0.060 |
| Region: Missing | 0.039 | 0.194 |

*Notes:* Unweighted means and standard deviations over 281,468 parent–offspring pairs. Incomes are permanent monthly wages in US dollars before taking logs. The first two rows are the variables the elasticity relates, and the remaining rows are the candidate predictors, measured on the parents. Where both parents are observed, age and years of education are averaged, the language indicator equals one if either parent speaks an indigenous language, and the categorical predictors, including gender, take the father's value. Years of education are missing for 3.0 per cent of pairs and the language indicator for 4.5 per cent, and their means are over the pairs that report them. For a categorical predictor, each row gives the share of pairs in one category.

## Variable Selection

Table E.6: Ecuador: Contribution Ratios

| Predictor | Mean $S$ | 2.5th pct | 97.5th pct |
|---|---|---|---|
| Indigenous language | 2.90 | 1.96 | 4.03 |
| Age | 1.89 | 1.18 | 2.88 |
| Years of education | 0.48 | 0.45 | 0.52 |
| Employment sector | 0.31 | 0.29 | 0.33 |
| Region | 0.05 | 0.03 | 0.07 |
| Gender | 0.04 | 0.01 | 0.08 |

*Notes:* Covariance-to-income contribution ratio $S_i(z_k)$ from Equation (32), computed against log offspring income for each candidate predictor. The screen removes a predictor whose ratio exceeds the cutoff $\tau = 10$. The percentiles come from 100 subsamples. No ratio comes close to the cutoff, and I retain all six predictors. These ratios are computed on simulated data rather than on the administrative records, so the screening is provisional.

## Regime Diagnostics

Table E.7: Ecuador: Regime Diagnostics

| Quantity | Subsamples | Full sample | Mean | 2.5th pct | 97.5th pct |
|---|---|---|---|---|---|
| $R^2$ | Joint | 0.648 | 0.648 | 0.641 | 0.654 |
| $\rho$ | Joint | 0.175 | 0.175 | 0.164 | 0.185 |
| $\rho^*$ | Joint | 0.152 | 0.152 | 0.147 | 0.158 |
| $\lambda$ | Joint | 0.868 | 0.872 | 0.815 | 0.938 |
| $\mathrm{Cov}(\tilde{\eta}, z)$ | Joint | 0.049 | 0.049 | 0.047 | 0.051 |
| $\mathrm{Cov}(\varepsilon, z)$ | Joint | 0.056 | 0.056 | 0.053 | 0.060 |
| $R^2$ | Split | | 0.649 | 0.642 | 0.655 |
| $\rho^*$ | Split | | 0.152 | 0.144 | 0.159 |
| $\mathrm{Cov}(\tilde{\eta}, z)$ | Split | | 0.049 | 0.046 | 0.051 |

*Notes:* Joint subsamples draw one sample of 30,000 pairs and use it as both donor and target, which $\rho$ and $\lambda$ require because they need both incomes for the same pairs. Split subsamples draw donor and target independently, the setting a researcher faces, where only $\rho^*$ can be computed. The percentiles come from 100 subsamples. Both incomes in logs.

## Quantile Gap and Imputed Distributions

Unlike those of the other applications, the quantile gap in Figure E.6 is not monotone. It rises to about 0.1 near the 12th percentile, falls to about −0.3 near the 29th, and only then rises steadily. Two features of the data produce this shape. Parents affiliated to the peasant social security regime make up 16 per cent of the sample and have far lower registered wages, with a median of about $200 a month, so their predicted incomes form a separate cluster at the bottom of the predicted distribution. Observed incomes outside that regime pile up in a band between about $325 and $425 a month, close to the minimum wage of the period.

Near the top of the peasant cluster, the observed quantiles have already reached that band while the predicted ones are still inside the cluster, so the gap is positive. Just above the cluster, the predicted quantiles jump to the rest of the distribution while the observed ones are still in the band, so the gap turns negative. The adjustment reproduces the observed shape, band included, whereas stochastic augmentation smooths it (Figure E.7). I show in Appendix A that the sign results do not depend on a monotone gap, and I discuss in Section 6.2 what the shape difference does to the correlation.

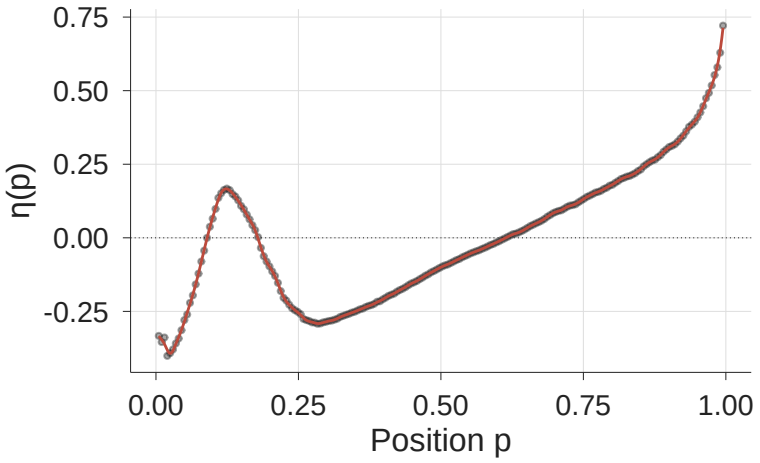



Figure E.6: Ecuador: Estimated Quantile Gap

*Notes:* Estimated quantile gap $\hat{\eta}(p)$ from a single replicate. Points give the gap at each of the 200 grid positions and the line is the smoothing spline. At each rank $p$, the gap is the amount the adjustment adds to predicted parental income, so negative values lower predicted income and positive values raise it. Income in logs.

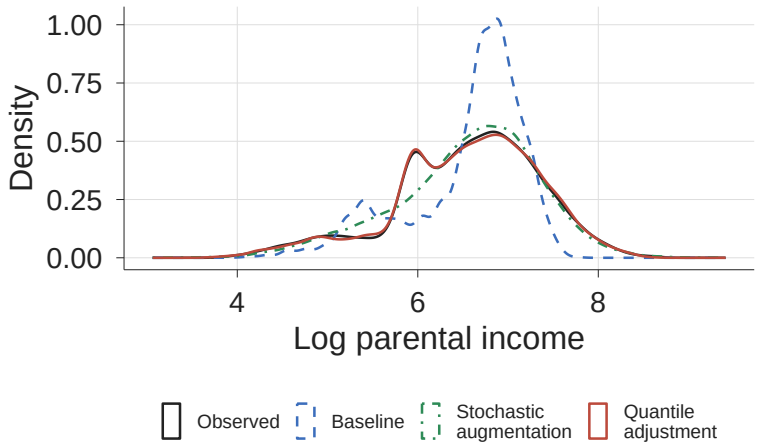



Figure E.7: Ecuador: Imputed and Observed Parental Income

*Notes:* Kernel density estimates of imputed log parental income in the target sample under each method, from the same replicate as Figure E.6, against the observed distribution. A method that restores the distribution of income produces a density close to the observed one.

## E.1 Rank-Rank Slope

The rank-rank slope regresses the offspring's percentile rank on the parent's percentile rank, a measure of mobility that does not depend on the scale of either income. I report it under the three methods in Table E.8 and the bias ratios in Figure E.8.

Baseline imputation understates the benchmark slope of 0.399 by 2.5 per cent, much less than it distorts the elasticity, because ranks remove the variance bias. The quantile adjustment returns the same slope. It is an increasing transformation of the prediction, so it leaves every target observation's rank unchanged, and a statistic built on ranks cannot tell it apart from baseline imputation.

The 2.5 per cent that remains is the part of the rank relationship the predictors miss, which only a better first stage could reduce. Stochastic augmentation adds noise unrelated to offspring income, which reshuffles parental ranks and lowers the slope by 24.1 per cent.

Table E.8: Ecuador: Rank-Rank Slope

| | Benchmark | Baseline | Stochastic | Quantile |
|---|---|---|---|---|
| Rank-rank slope | 0.399 | 0.389<br>[0.379, 0.398] | 0.303<br>[0.292, 0.313] | 0.389<br>[0.379, 0.398] |
| Relative bias (%) | | −2.5 | −24.1 | −2.5 |
| Relative RMSE (%) | | 2.8 | 24.1 | 2.8 |

*Notes:* Benchmark value from the full working sample. Brackets give the 2.5th and 97.5th percentiles across 1,000 replicates of the split design described in Section 5. Ranks are percentile ranks of parental and offspring income.

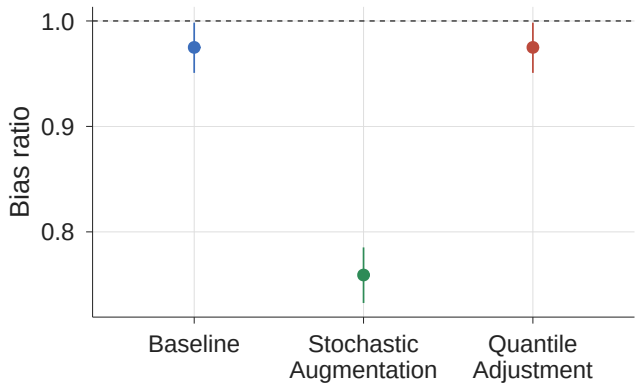



Figure E.8: Ecuador: Bias Ratio for the Rank-Rank Slope

*Notes:* Bias ratio for the rank-rank slope under the three imputation methods. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th percentiles across 1,000 replicates of the split design described in Section 5.

# F Regressions and Inequality Measures: Supplementary Material

Here I collect supplementary material for the inequality of opportunity application of Section 5.3, which regresses household income on four circumstances. Appendix F.1 repeats the exercise without the screen, and I use that comparison in Section 6.3.

## Working Sample

Table F.9: Chile: Working Sample

| Variable | Mean | SD | Variable | Mean | SD |
|---|---|---|---|---|---|
| Log income | 12.524 | 0.824 | Region: 5 | 0.102 | 0.303 |
| Male | 0.433 | 0.496 | Region: 6 | 0.055 | 0.228 |
| Indigenous | 0.078 | 0.269 | Region: 7 | 0.060 | 0.237 |
| Father's education | 0.110 | 0.312 | Region: 8 | 0.112 | 0.316 |
| Mother's education | 0.081 | 0.273 | Region: 9 | 0.056 | 0.231 |
| Household size | 3.603 | 1.726 | Region: 10 | 0.051 | 0.219 |
| Share male | 0.501 | 0.232 | Region: 11 | 0.006 | 0.074 |
| Share occupied | 0.488 | 0.299 | Region: 12 | 0.009 | 0.093 |
| Education of head | 11.251 | 4.442 | Region: 13 | 0.408 | 0.491 |
| Region: 1 | 0.018 | 0.133 | Region: 14 | 0.024 | 0.152 |
| Region: 2 | 0.029 | 0.168 | Region: 15 | 0.011 | 0.104 |
| Region: 3 | 0.018 | 0.133 | Gender of head: 1 | 0.713 | 0.452 |
| Region: 4 | 0.042 | 0.200 | Gender of head: 2 | 0.287 | 0.452 |

*Notes:* Weighted means and standard deviations over 98,950 observations, representing 6,704,013 people. The first row is the income variable, the next four are the circumstances, and the rest are the candidate predictors. The circumstances are binary. Father's and mother's education equal one when the parent completed secondary education or above. For a categorical predictor, each row gives the share of observations in one category.

## Variable Selection

Table F.10 reports the two predictors the screen removes and the round at which it removes them, and Table F.11 the ratios of the four that survive.

Table F.10: Chile: Predictors Removed by the Screen

| Round | Predictor | Flagged against | Mean $S$ | [2.5th, 97.5th pct] |
|---|---|---|---|---|
| 1 | Share male | Male | $6.9 \times 10^7$ | [159, $3.1 \times 10^6$] |
| 2 | Gender of head | Male | 54.6 | [32.4, 94.4] |

*Notes:* Covariance-to-income contribution ratio $S_i(z_k)$ from Equation (32) for each predictor in the round at which the screen removed it, against the circumstance that flagged it. I recompute the ratios after each removal, so the second value assumes the share of male members is already out. The percentiles come from 100 subsamples, a few of which produce enormous ratios that pull the mean above its own 97.5th percentile.

Table F.11: Chile: Contribution Ratios, Surviving Predictors

| Predictor | Male | Indigenous | Father's education | Mother's education |
|---|---|---|---|---|
| Region | 0.04 | 2.38 | 0.30 | 0.22 |
| Education of head | 0.01 | 0.05 | 0.71 | 0.52 |
| Share occupied | 0.12 | 0.00 | 0.00 | 0.02 |
| Household size | 0.03 | 0.01 | 0.01 | 0.01 |

*Notes:* Covariance-to-income contribution ratio $S_i(z_k)$ from Equation (32) against each circumstance, computed on the final specification. The screen removes a predictor when any of its ratios exceeds the cutoff $\tau = 10$, and none does here. The replication files report percentiles across subsamples.

## Regime Diagnostics

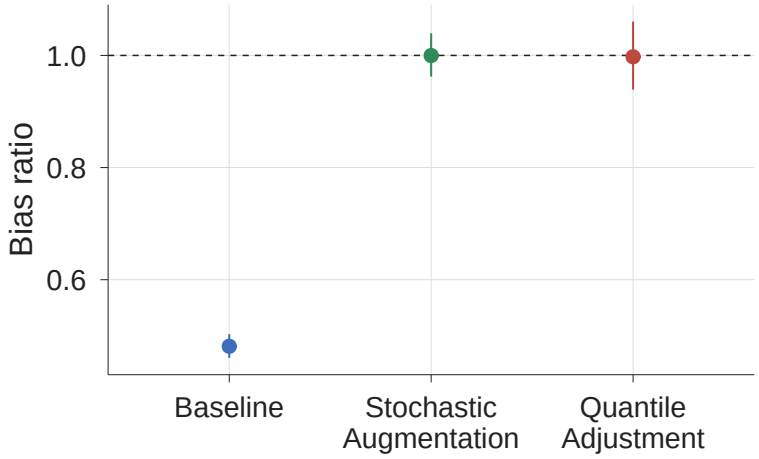
Table F.12: Chile: Regime Diagnostics

| Circumstance | $\rho^*$ | $\rho$ | $\lambda$ | [2.5th, 97.5th pct] | Regime |
|---|---|---|---|---|---|
| Male | 0.056 | 0.044 | 1.282 | [0.859, 2.218] | Over-recovery |
| Indigenous | −0.034 | −0.064 | 0.539 | [0.426, 0.764] | Partial covariance adjustment |
| Father's education | 0.141 | 0.172 | 0.819 | [0.699, 0.960] | Partial covariance adjustment |
| Mother's education | 0.130 | 0.132 | 0.986 | [0.819, 1.222] | Partial covariance adjustment |

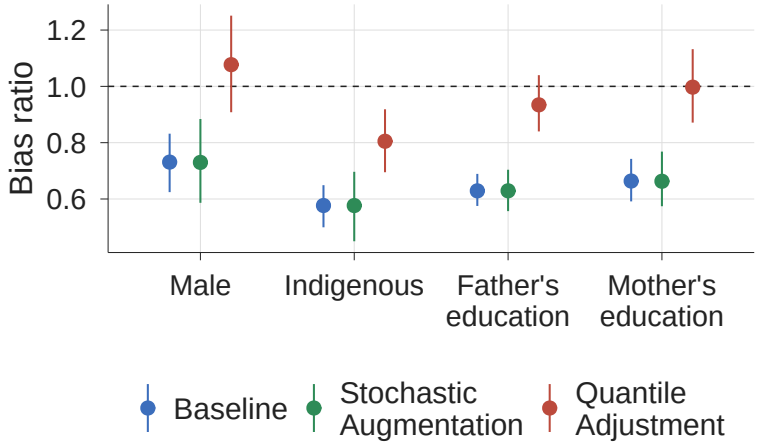
*Notes:* $\rho^*$, $\rho$ and $\lambda$ come from the full working sample, which observes income and the circumstances jointly, and the first stage attains $R^2 = 0.481$. The percentiles come from 100 subsamples, each serving as both donor and target, which $\rho$ and $\lambda$ require. The intervals for the respondent's sex and mother's education cross a regime boundary.

## Variance and Covariances

The circumstance coefficients of Table 5 are built from the variance of log income and its covariance with each circumstance. Figure F.9 gives the bias ratios for those moments.




(a) Var($y$)




(b) Cov($y, z_j$)

Figure F.9: Chile: Bias Ratios for the Variance and Covariances

*Notes:* Bias ratio for the variance of log income and for its covariance with each circumstance, under the three imputation methods. These are the moments the circumstance coefficients are built from. The horizontal dashed line marks one, alignment with the benchmark. Points give the mean and bars the 2.5th and 97.5th percentiles across 1,000 replicates of the split design described in Section 5. Income in logs.

## Quantile Gap and Imputed Distributions

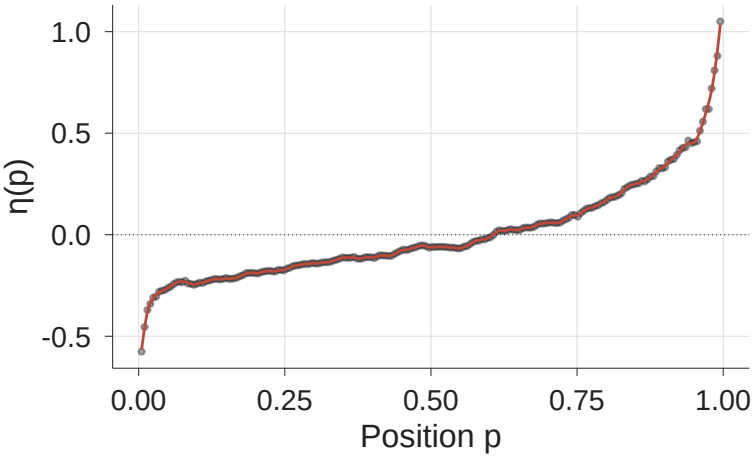



Figure F.10: Chile: Estimated Quantile Gap

*Notes:* Estimated quantile gap $\hat{\eta}(p)$ from a single replicate. Points give the gap at each of the 200 grid positions and the line is the smoothing spline. At each rank $p$, the gap is the amount the adjustment adds to predicted income, so negative values lower predicted income and positive values raise it. Income in logs.

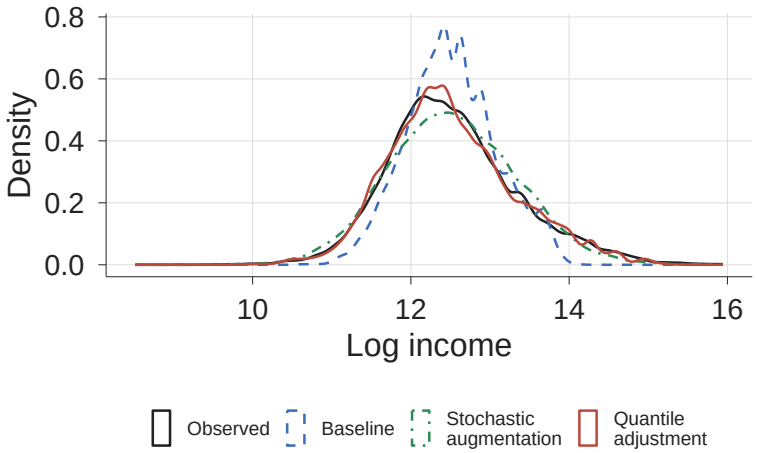



Figure F.11: Chile: Imputed and Observed Income

*Notes:* Kernel density estimates of imputed log income in the target sample under each method, from the same replicate as Figure F.10, against the observed distribution. A method that restores the distribution of income produces a density close to the observed one.

## F.1 Unscreened Specification

Here I ask what the screen changes, by repeating the Chilean exercise with all six candidate predictors and keeping the share of male household members and the gender of the household head that the screen of Section 4.4 removes. Everything else is unchanged. I compare the recovery fractions with and without the screen in Table F.13 and report the estimates in Table F.14, and I discuss both in Section 6.3.

Table F.13: Chile: Recovery Fractions With and Without the Screen

| | Screened | Unscreened |
|---|---|---|
| $R^2$ | 0.481<br>[0.468, 0.494] | 0.483<br>[0.470, 0.496] |
| $\lambda_{\text{Male}}$ | 1.340<br>[0.884, 2.080] | 2.925<br>[1.625, 5.252] |
| $\lambda_{\text{Indigenous}}$ | 0.543<br>[0.417, 0.692] | 0.538<br>[0.411, 0.683] |
| $\lambda_{\text{Father's education}}$ | 0.828<br>[0.663, 1.023] | 0.811<br>[0.656, 1.003] |
| $\lambda_{\text{Mother's education}}$ | 1.002<br>[0.759, 1.329] | 0.965<br>[0.733, 1.290] |

*Notes:* $\lambda_j$ is the share of the covariance omitted for circumstance $j$ that the adjustment recovers, computed within each replicate. Brackets give the 2.5th and 97.5th percentiles across 1,000 replicates of the split design described in Section 5. The unscreened specification keeps the share of male household members and the gender of the household head, and the other three recovery fractions barely change.

Table F.14: Chile: Unscreened Estimates

| | Benchmark | Baseline | Stochastic | Quantile |
|---|---|---|---|---|
| *Panel A. Log scale* | | | | |
| $\mathrm{Var}(y)$ | 0.678 | 0.328<br>[0.314, 0.343] | 0.678<br>[0.653, 0.705] | 0.675<br>[0.635, 0.718] |
| $\mathrm{Cov}(y, \mathrm{Male})$ | 0.048 | 0.040<br>[0.035, 0.046] | 0.040<br>[0.033, 0.048] | 0.060<br>[0.051, 0.068] |
| $\mathrm{Cov}(y, \mathrm{Indigenous})$ | −0.023 | −0.013<br>[−0.015, −0.012] | −0.013<br>[−0.016, −0.011] | −0.019<br>[−0.021, −0.016] |
| $\mathrm{Cov}(y, \mathrm{Father's\ education})$ | 0.081 | 0.051<br>[0.046, 0.055] | 0.051<br>[0.045, 0.057] | 0.075<br>[0.067, 0.083] |
| $\mathrm{Cov}(y, \mathrm{Mother's\ education})$ | 0.061 | 0.040<br>[0.036, 0.045] | 0.040<br>[0.035, 0.046] | 0.060<br>[0.052, 0.068] |
| $\delta_{\mathrm{Male}}$ | 0.177 | 0.154<br>[0.135, 0.172] | 0.153<br>[0.126, 0.181] | 0.227<br>[0.197, 0.257] |
| $\delta_{\mathrm{Indigenous}}$ | −0.257 | −0.143<br>[−0.167, −0.119] | −0.143<br>[−0.182, −0.103] | −0.195<br>[−0.230, −0.160] |
| $\delta_{\mathrm{Father's\ education}}$ | 0.610 | 0.363<br>[0.328, 0.397] | 0.364<br>[0.312, 0.422] | 0.532<br>[0.473, 0.591] |
| $\delta_{\mathrm{Mother's\ education}}$ | 0.396 | 0.286<br>[0.246, 0.327] | 0.285<br>[0.217, 0.355] | 0.435<br>[0.365, 0.505] |
| *Panel B. Level scale* | | | | |
| Total inequality $I(y)$ | 0.482 | 0.320<br>[0.313, 0.327] | 0.451<br>[0.440, 0.464] | 0.477<br>[0.459, 0.497] |
| Absolute IOp $I(y_{\mid \mathbf{z}})$ | 0.161 | 0.101<br>[0.094, 0.109] | 0.102<br>[0.091, 0.113] | 0.158<br>[0.144, 0.175] |
| Relative IOp $\theta$ | 0.334 | 0.317<br>[0.296, 0.339] | 0.225<br>[0.203, 0.247] | 0.332<br>[0.307, 0.358] |

*Notes:* As Table 5, with the share of male household members and the gender of the household head kept in the first stage. Compared with the screened specification, the covariance and coefficient for the respondent's sex move further above the benchmark, while the relative share and absolute IOp come slightly closer to it.

# G Bias and Variation Across Replicates

The relative RMSE of Table 6 combines two sources of error. The mean squared deviation from the benchmark equals the squared bias, the part that repeats in every replicate, plus the variance across replicates, the part that averages out. I report the first as a share of the total in Table G.15.

A share near 100 per cent means that almost all of the error is bias, which a larger sample would not remove. A share near zero means that the method is right on average and its error comes from sampling variation. I use the table in Section 6.1 to compare the methods.

Table G.15: Share of Error Due to Bias (%)

| | Baseline | Stochastic | Quantile |
|---|---|---|---|
| *Mexico* | | | |
| $\mathrm{Var}(y)$ | 100.0 | 0.0 | 9.6 |
| $\mathrm{Cov}(y, z)$ | 99.9 | 99.9 | 99.2 |
| $\rho_{y,z}$ | 99.8 | 99.9 | 99.8 |
| *Ecuador* | | | |
| $\mathrm{Var}(y)$ | 99.9 | 0.0 | 0.0 |
| $\mathrm{Cov}(y, z)$ | 99.7 | 99.5 | 77.4 |
| $\rho_{y,z}$ | 86.9 | 99.7 | 86.8 |
| $\beta_{\mathrm{IGE}}$ | 99.4 | 99.6 | 84.3 |
| *Chile* | | | |
| $\mathrm{Var}(y)$ | 100.0 | 0.0 | 0.6 |
| $\mathrm{Cov}(y, \mathrm{Male})$ | 96.3 | 93.0 | 44.9 |
| $\mathrm{Cov}(y, \mathrm{Indig.})$ | 99.2 | 97.9 | 92.2 |
| $\mathrm{Cov}(y, \mathrm{Father})$ | 99.4 | 98.9 | 60.4 |
| $\mathrm{Cov}(y, \mathrm{Mother})$ | 98.7 | 97.9 | 0.2 |
| $\delta_{\mathrm{Male}}$ | 96.4 | 92.2 | 54.7 |
| $\delta_{\mathrm{Indig.}}$ | 98.8 | 97.1 | 92.0 |
| $\delta_{\mathrm{Father}}$ | 99.5 | 98.7 | 87.1 |
| $\delta_{\mathrm{Mother}}$ | 96.3 | 90.6 | 64.0 |
| $I(y)$ | 100.0 | 96.4 | 9.5 |
| $I(y_{\mid \mathbf{z}})$ | 99.6 | 99.2 | 39.4 |
| $\theta$ | 86.3 | 99.1 | 41.8 |

*Notes:* Squared bias as a share of the mean squared deviation from the benchmark under each method, computed across 1,000 replicates of the split design described in Section 5. The remainder is variation across replicates. Rows match Table 6.